

Asana: Real-Time Yoga Pose Analysis with Rust

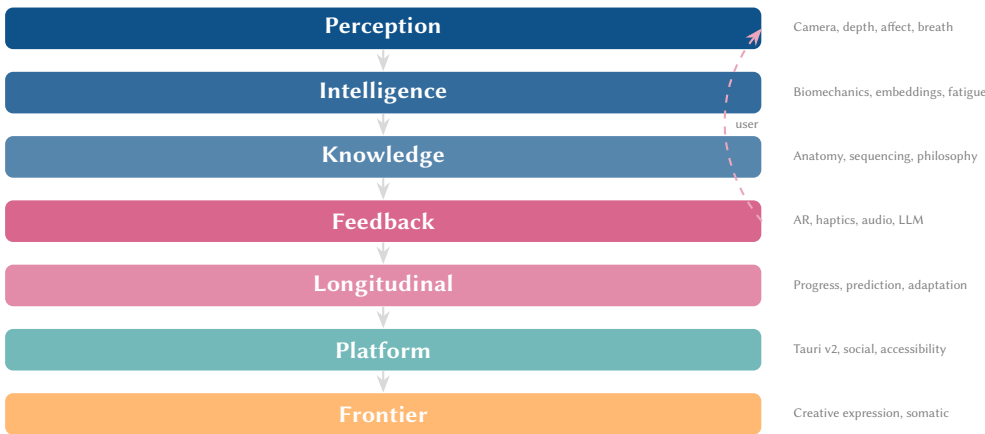
Architecture and Feasibility Study for an Intelligent Practice Companion

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Abstract. We present *Asana*, a comprehensive architecture for a Rust-based yoga practice companion that spans real-time pose estimation, biomechanical analysis, domain knowledge modelling, and longitudinal practice intelligence. Starting from monocular webcam input processed by RTMPose via ONNX Runtime, the system builds seven successive layers of capability: (1) a **perception layer** fusing camera, depth, facial-affect, and breath signals; (2) an **intelligence layer** deriving joint torques, balance metrics, fatigue indices, and pose embeddings from raw keypoints; (3) a **knowledge layer** encoding myofascial meridians, sequencing constraints, and pose genealogy; (4) a **feedback layer** delivering corrections through AR overlay, haptic wearables, audio cueing, and LLM-powered explanation; (5) a **longitudinal layer** tracking progress, predicting milestones, and adapting difficulty over months of practice; (6) a **platform layer** providing the Tauri v2 application shell, social features, accessibility, and privacy-preserving analytics; and (7) a **frontier layer** exploring creative expression, dream-sequence reconstruction, and neuroplasticity-informed motor learning. We evaluate each layer’s feasibility on commodity hardware and demonstrate that real-time pose feedback at 30 fps is achievable within a 33 ms per-frame budget using RTMPose-m, while the extended feature set maps cleanly onto the proposed Tauri v2 + Rust backend architecture with a clear incremental development path.

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ONNX	Open Neural Network Exchange	3
DTW	Dynamic Time Warping	26
SBL	Superficial Back Line	18
SFL	Superficial Front Line	18
DFL	Deep Front Line	18
AP	Average Precision	39
COCO	Common Objects in Context	3
IPC	Inter-Process Communication	5
SimCC	Simulated Coordinate Classification	3
EMA	Exponential Moving Average	5
kNN	k-Nearest Neighbours	8
SVD	Singular Value Decomposition	8

VISION AND PRODUCT THESIS

This part frames the product vision and architectural approach that motivates the rest of the report. We begin with the problem statement and layered architecture (section 1), then outline the development roadmap that will unfold across the subsequent six parts.

1 Introduction and Motivation

Yoga practice benefits enormously from feedback on alignment, timing, and sequencing—traditionally the province of an experienced instructor. The convergence of real-time pose estimation models, efficient ML inference runtimes, and cross-platform Rust tooling makes it feasible to build a software companion that provides this feedback autonomously. But alignment correction is only the beginning: a truly intelligent practice companion must *sense* the practitioner, *understand* their movement, draw on *domain knowledge*, *respond* with appropriate feedback, *learn* from longitudinal data, *scale* across platforms and communities, and *explore* creative frontiers.

Insight:
A monocular camera provides sufficient signal to extract actionable alignment corrections for most standing and seated poses.

This report develops each of these capabilities in turn, organised into seven architectural layers (fig. 1):

1. **Perception** (part II)—camera capture, pose estimation, depth sensing, facial affect, and breath signals.
2. **Intelligence** (part III)—deriving biomechanical quantities, pose embeddings, fatigue indices, and anomaly flags from raw keypoints.
3. **Knowledge** (part IV)—encoding yoga domain expertise: anatomy, sequencing, philosophy, and circadian context.
4. **Feedback** (part V)—delivering corrections and insights via AR overlay, audio cueing, haptics, and natural language.
5. **Longitudinal Intelligence** (part VI)—tracking progress, predicting milestones, and adapting practice over time.
6. **Platform and Ecosystem** (part VII)—the Tauri v2 application shell, social features, accessibility, and privacy-preserving analytics.
7. **Creative and Frontier** (part VIII)—speculative features exploring the boundaries of movement technology.

PERCEPTION LAYER

With the architectural vision established, we turn to the system’s first responsibility: *sensing the practitioner*. This part evaluates pose estimation models, the Rust inference ecosystem, camera and depth capture, and supplementary signals from facial expression, breath sound, and props.

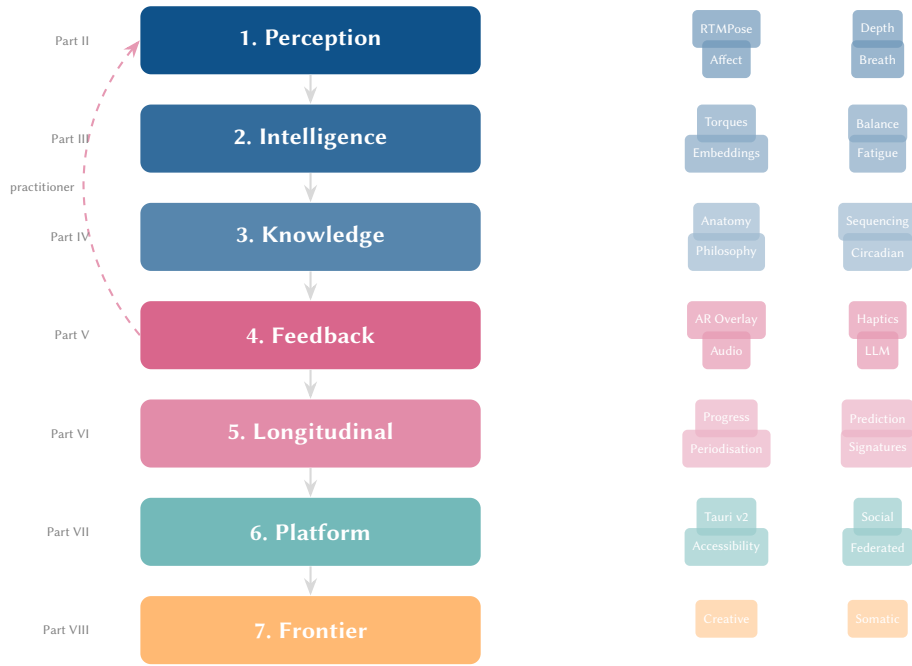


Figure 1. Seven-layer architecture of the Asana system. Each layer is developed in its corresponding Part of this report. Data flows downward from perception through intelligence, knowledge, and feedback; the feedback loop returns corrections to the practitioner. Longitudinal, platform, and frontier layers wrap the core sensing–response loop.

2 Pose Estimation Model Landscape

Six families of pose estimation models were evaluated against criteria relevant to a yoga application: keypoint count, accuracy on Common Objects in Context (COCO) val, inference latency on both GPU and CPU, mobile viability, and licence compatibility.

2.1 Model Comparison

Table 1. Pose estimation model comparison. Latency measured at native resolution. [AP](#) on COCO val2017.

Model	Keypoints	AP	GPU (ms)	CPU (ms)	Mobile	Licence
BlazePose [1]	33	~70*	5–8	15–30	✓	Apache 2.0
MoveNet Lightning [2]	17	70.5	4–6	12–15	✓	Apache 2.0
MoveNet Thunder [2]	17	72.3	8–11	25–30	✓	Apache 2.0
OpenPose [3]	25	61–65	30–50	200–500	×	Non-comm.
YOLOv8m-pose [4]	17	71.5	5	60–100	Partial	AGPL-3.0
RTMPose-m [5]	17	75.3	3.5	15–20	✓	Apache 2.0
RTMPose-l [5]	17	76.3	5	25–30	Partial	Apache 2.0
ViTPose-B [6]	17	75.8	15–25	N/A	×	Apache 2.0
ViTPose-H [6]	17	79.1	40–60	N/A	×	Apache 2.0

*BlazePose [AP](#) not directly comparable due to different keypoint definition (33 vs. 17).

: Model Selection

RTMPose-m is the primary recommendation: 75.3 [AP](#) at 3.5 ms GPU / 15–20 ms CPU, Apache 2.0 licence, clean [ONNX](#) export via MMDeploy, and a [SimCC](#) head that eliminates argmax/heatmap post-processing from the model graph.

Insight:
RTMPose’s SimCC head avoids expensive heatmap post-processing, making it ideal for ONNX deployment.

2.2 Architecture Characteristics

RTMPose [5] uses a CSPNeXt backbone with a Simulated Coordinate Classification ([SimCC](#)) head that formulates keypoint localization as a classification problem along each coordinate axis independently. This avoids the heatmap decoding step required by most other models, yielding both faster inference and simpler Open Neural Network Exchange ([ONNX](#)) graph topology.

For yoga specifically, **BlazePose** [1] offers a compelling alternative due to its 33 keypoints—including heel, foot index, and hand landmarks—which are critical for detecting foot placement in standing poses and hand

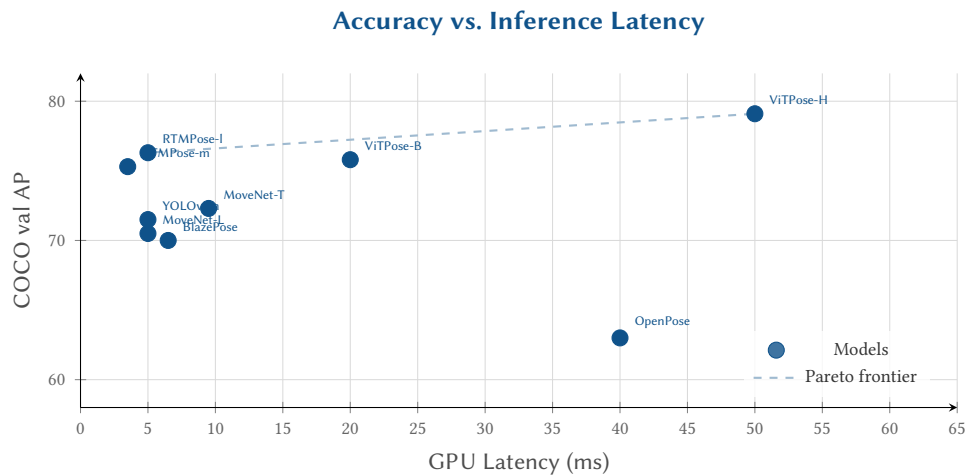


Figure 2. Accuracy–latency Pareto analysis of pose estimation models. RTMPose variants dominate the real-time region.

positioning in binds. The trade-off is lower per-keypoint accuracy and a more complex deployment path (TFLite-native models requiring conversion).

Yoga-Specific Keypoint Needs

Standard COCO 17-keypoint models lack foot detail (only ankle) and hand detail (only wrist). For poses like Warrior I (back foot angle), Tree Pose (foot placement on inner thigh), or Crow Pose (hand width), supplementary hand/foot keypoint models or BlazePose’s 33-point superset may be required.

3 Rust ML Inference Ecosystem

Seven Rust-based approaches to ML inference were evaluated. Table 2 summarises the key characteristics.

Table 2. Rust ML inference runtime comparison.

Crate	GPU	Pure Rust	ONNX	Maturity	Notes
ort v2.x [7]	✓	×	✓	High	Recommended. CoreML, CUDA, TensorRT.
tract v0.21 [8]	×	✓	✓	High	Best pure-Rust option. SIMD-optimised CPU.
burn v0.15	wgpu	✓	Partial	Medium	ONNX import incomplete for pose models.
candle v0.8	CUDA/Metal	✓	×	Medium	No ONNX import; must reimplement models.
tch-rs v0.17	✓	×	×	Med-Hi	Requires 2 GB libtorch dependency.
tflitec v0.5	Partial	×	×	Low	TFLite models only. Delegate support weak.
MediaPipe FFI	✓	×	×	N/A	Bazel build; no official Rust bindings.

: Inference Runtime

Use ort v2.x (ONNX Runtime) as the primary inference backend. It provides the broadest execution provider coverage (CoreML on macOS for Apple Neural Engine acceleration, CUDA/TensorRT on NVIDIA, DirectML on Windows) with thin, well-maintained Rust bindings. For a pure-Rust fallback, tract provides competitive CPU performance with zero C dependencies.

3.1 Inference Pipeline

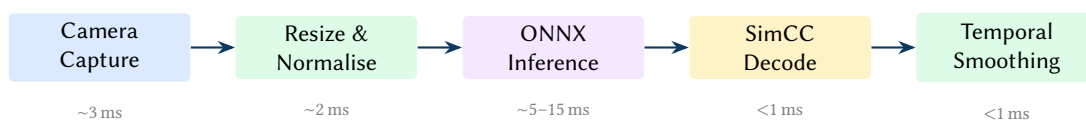


Figure 3. Per-frame inference pipeline with target latency budget (total < 33 ms for 30 fps).

The pipeline operates within a 33 ms frame budget:

1. **Camera Capture** via nokhwa [9] (cross-platform: AVFoundation on macOS, V4L2 on Linux, MediaFoundation on Windows).

Insight:
ort v2 can auto-download ONNX Runtime binaries via the download-binaries feature flag, simplifying builds.

2. **Preprocessing:** resize to model input (256×192), BGR→RGB conversion, mean/std normalisation. Uses the image crate for pure-Rust processing.
3. **Inference:** RTMPose-m via ort with platform-appropriate execution provider.
4. **SimCC Decode:** extract (x, y) coordinates from classification logits—a simple argmax per axis.
5. **Temporal Smoothing:** Exponential Moving Average (EMA) over 3–5 frames to reduce jitter ($\alpha = 0.6-0.8$).

4 Camera Capture

Table 3. Camera capture options in Rust.

Approach	macOS	Linux	Windows	Notes
nokhwa v0.10	✓	✓	✓	Recommended; cross-platform
opencv VideoCapture	✓	✓	✓	Heavy dependency (~20 min build)
v4l	×	✓	×	Linux-specific low-level API
Browser getUserMedia	Via Tauri webview			Simplest for Tauri architecture

For the recommended Tauri architecture (section 58), camera capture can be handled entirely in the browser layer via `getUserMedia`, avoiding the need to shuttle raw frame data across the Inter-Process Communication (IPC) boundary. The Rust backend receives only keypoint coordinates (~100 bytes/frame), not megabytes of video.

Design Decision 1: Camera Strategy

Two viable strategies exist: (1) browser-side capture with `getUserMedia` feeding a `<video>` element, with frames extracted to a `<canvas>` and sent to the Rust backend for inference; or (2) Rust-side capture via `nokhwa` with the backend returning keypoints to the frontend for overlay rendering. Strategy (2) keeps ML inference entirely in Rust but requires frame data to cross the IPC boundary at least once.

5 Multi-Camera and Depth Sensing

A single monocular camera cannot reliably assess all alignment errors. Multi-camera or depth-sensing approaches substantially reduce blind spots.

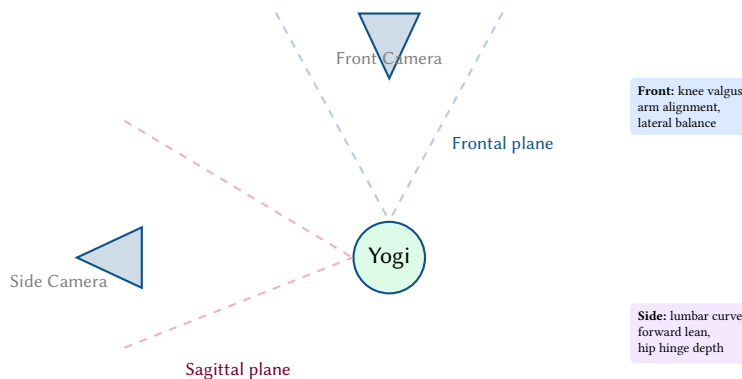


Figure 4. Dual-camera setup: front camera captures frontal-plane errors, side camera captures sagittal-plane errors. Combined, they cover the full range of common alignment issues.

Design Decision 2: Progressive Camera Support

Phase 1 targets single-camera use. The system guides the user to the optimal camera angle per pose category. Phase 2 adds optional dual-camera support (phone + laptop, or two phones) with automatic triangulation. Future phases could integrate depth cameras (Apple TrueDepth, Intel RealSense, Azure Kinect) for full 3D reconstruction.

6 Props and Surface Detection

Using a lightweight segmentation model on keyframes (every 5 s), the system identifies the practice surface and available props.

Table 4. Detectable props and their effect on cue generation.

Detected Item	Cue Adjustment	Example
Yoga block	Suggest block use for limited ROM	“Place block under hand in Triangle”
Strap	Suggest strap for binds/reaches	“Loop strap around foot in Dancer”
Bolster	Enable restorative pose variants	“Place bolster under knees in Savasana”
Wall	Suggest wall-supported inversions	“Try L-shape handstand against wall”
Chair	Activate chair yoga mode	Auto-switch to seated variants
Foam roller	Offer pre-practice release protocol	“Roll IT band before standing poses”

Design Decision 3: Surface Compliance Estimation

The practice surface affects safety: hard floors increase joint impact, slippery surfaces affect BoS stability, grass adds proprioceptive challenge. A MobileNetV3 segmentation head (~5 MB) classifies the floor region under the practitioner. Cue adjustments include: extra padding suggestions for hard floors, grip cues for slippery surfaces, and longer warm-ups on cold surfaces.

7 Facial Affect and Effort Recognition

Beyond body pose, the camera captures facial expressions that encode effort, discomfort, and concentration—signals invisible to a skeleton-only system.

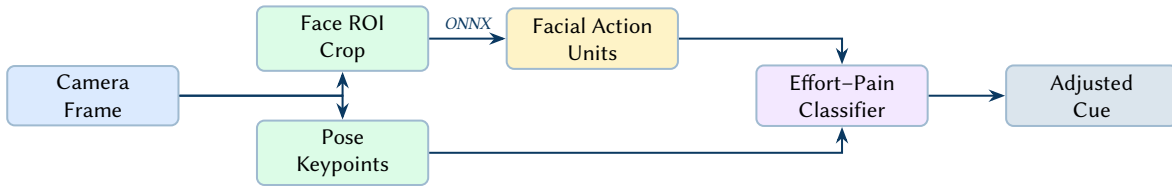


Figure 5. Facial affect pipeline: a lightweight facial action unit model distinguishes productive effort from pain, adjusting the correction strategy accordingly.

- **Effort vs. pain discrimination:** if the face shows strain while the pose looks “correct,” surface a gentler cue (breathing reminder) rather than a technical adjustment.
- **Concentration tracking:** sustained gaze stability and reduced blink rate correlate with meditative focus; increased AU4 (brow lowerer) correlates with frustration.
- **Enjoyment detection:** genuine smiles (Duchenne marker: AU6 + AU12) during practice correlate with intrinsic motivation and session adherence.

8 Vocal Breath and Sound Analysis

Microphone input adds an acoustic channel complementing the shoulder-oscillation breath estimation (section 16).

1. **Ujjayi detection:** the characteristic “ocean sound” of ujjayi breath has a distinctive spectral signature (broadband turbulent noise centred at 200–600 Hz). A small CNN on STFT features classifies breathing style.
2. **Breath holding detection:** absence of both acoustic and shoulder-movement breath signals for >5 s during a pose indicates breath holding—a common beginner error that increases blood pressure and reduces muscle oxygenation.
3. **Exertion vocalisation:** involuntary grunting or sighing correlates with effort exceeding capacity. The system can suggest a modification before the practitioner consciously recognises they are struggling.
4. **Pranayama coaching:** for techniques like Kapalabhati (rapid abdominal breathing) or Nadi Shodhana (alternate nostril), acoustic analysis provides more precise feedback than shoulder movement alone.

9 Performance Budget

The backend-driven approach yields a total estimated latency of ~18.5 ms, providing 14.5 ms of headroom within the 33 ms frame budget. This headroom accommodates:

- Occasional inference spikes on CPU-only systems.
- Additional pose analysis computation (angle extraction, classification, correction lookup).

Insight:
The distinction between “growth edge” (productive challenge) and “pain” (injury risk) is visible on the face long before it manifests in the skeleton. A 50 KB facial action unit model adds negligible compute but transforms correction quality.

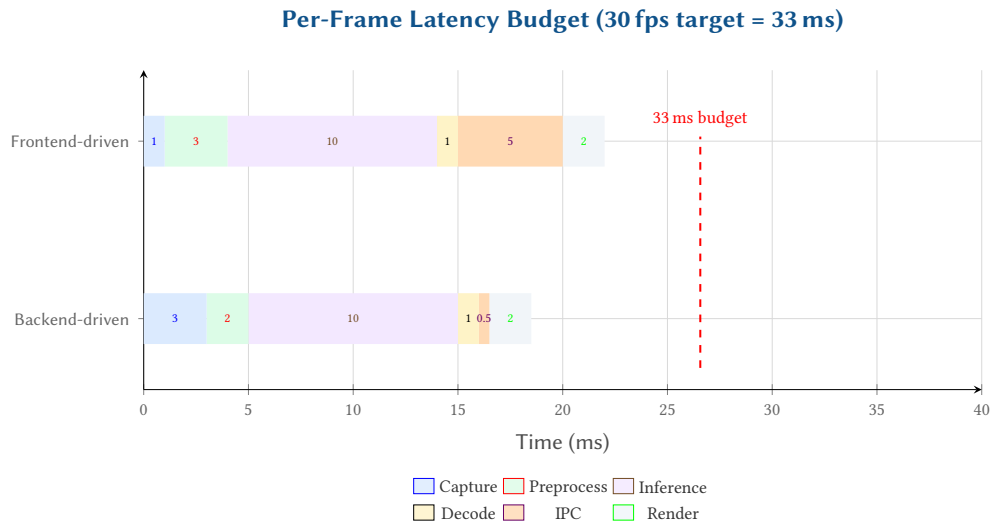


Figure 6. Latency breakdown for both data flow patterns. Backend-driven capture fits comfortably within the 33 ms budget.

- Frame-to-frame variability in camera capture timing.

INTELLIGENCE LAYER

Raw keypoints from the perception layer are necessary but not sufficient—they must be transformed into *meaningful* biomechanical and movement-quality signals. This part develops the analysis pipeline: from joint angles and corrections through inverse dynamics, balance, fatigue, pose embeddings, and anomaly detection.

10 From Keypoints to Corrections

Raw keypoint coordinates must be transformed into actionable feedback.

10.1 Joint Angle Calculation

Given three keypoints A , B (joint vertex), C defining a joint, the 2D angle is:

$$\theta = \arctan\left(\frac{C_y - B_y}{C_x - B_x}\right) - \arctan\left(\frac{A_y - B_y}{A_x - B_x}\right)$$

For 3D when depth estimates are available (BlazePose’s z -coordinate):

$$\theta = \arccos\left(\frac{(\vec{BA}) \cdot (\vec{BC})}{\|\vec{BA}\| \cdot \|\vec{BC}\|}\right)$$

Insight:
A hybrid approach—landmark embedding for pose classification, joint angles for correction—provides both “what pose is this?” and “how should I adjust?”

Table 5. Key joint angles for yoga pose analysis. Keypoint indices follow the COCO 17-point convention.

Joint	Keypoints	COCO Indices	Yoga Relevance
Left knee	hip–knee–ankle	11–13–15	Warriors, lunges
Right knee	hip–knee–ankle	12–14–16	Warriors, lunges
Left hip	shoulder–hip–knee	5–11–13	Forward folds
Right hip	shoulder–hip–knee	6–12–14	Forward folds
Left elbow	shoulder–elbow–wrist	5–7–9	Plank, Chaturanga
Right elbow	shoulder–elbow–wrist	6–8–10	Plank, Chaturanga
Left shoulder	elbow–shoulder–hip	7–5–11	Overhead poses
Right shoulder	elbow–shoulder–hip	8–6–12	Overhead poses
Torso incline	mid(shoulders)–mid(hips) vs. vertical	—	Standing alignment

10.2 Pose Classification

Pose identification uses a two-stage approach:

1. **Normalise** the skeleton: translate to hip centroid, scale by torso length, optionally Procrustes-align orientation via Singular Value Decomposition (SVD) [10] (using `nalgebra`).
2. **Classify** via k-Nearest Neighbours (kNN) on normalised landmark vectors against a reference database of ~50–60 canonical poses.

10.3 Correction Feedback Pipeline

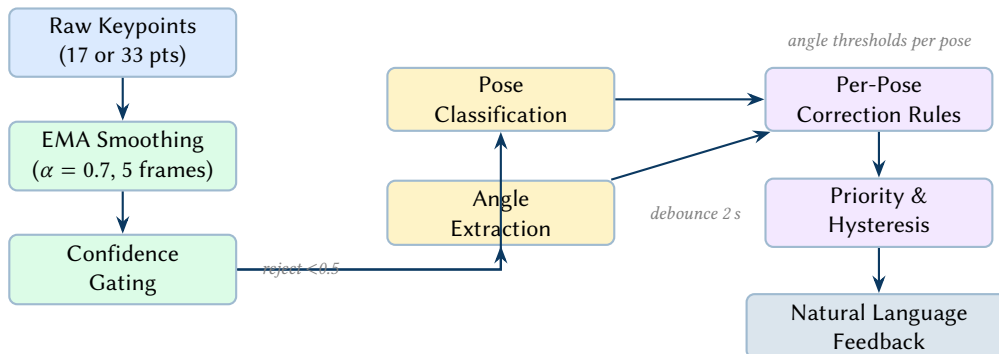


Figure 7. Feedback generation pipeline from raw keypoints to natural-language correction cues.

: Temporal Stability

Never flip-flop feedback between frames. Use hysteresis: a correction triggers only when the deviation persists for N frames ($N \geq 10$ at 30 fps = 330 ms). Once triggered, the same correction is not repeated for M seconds ($M \geq 3$). This prevents the jarring experience of rapidly changing instructions.

10.4 Handling 2D Limitations

A monocular camera introduces foreshortening and self-occlusion ambiguities. Mitigation strategies:

- **Confidence gating:** suppress feedback for joints with model confidence < 0.5 .
- **Camera angle guidance:** prompt the user to stand at $\sim 45^\circ$ for most poses; suggest “stand sideways” for sagittal-plane poses (Warrior III, Triangle).
- **Pseudo-depth:** if using BlazePose (33 keypoints with z -coordinate), use the estimated depth for 3D angle calculation on critical joints.
- **Pose-specific rules:** only evaluate joints that are reliably visible from the recommended camera angle.

11 Common Alignment Errors

Table 6 catalogues the most common alignment errors across 15 frequently practised poses, the biomechanical risk of each error, and the corrective cue. These form the initial rule base for the correction engine.

Table 6. Common alignment errors, risks, and correction cues for 15 yoga poses.

Pose	Error	Risk	Correction Cue
Downward Dog	Rounded upper back, shoulders near ears	Shoulder impingement	“Press chest toward thighs; draw shoulder blades toward hips”
Downward Dog	Hyperextended elbows	Elbow joint stress	“Maintain a micro-bend in the elbows”
Warrior II	Front knee collapsing inward (valgus)	MCL strain	“Track the front knee over the second toe; engage outer hip”
Warrior II	Torso leaning over front thigh	Quad overload	“Stack torso directly over pelvis”
Triangle	Chest rotating toward floor	Shoulder impingement	“Rotate chest open toward the sky”
Chaturanga	Shoulders dropping below elbows	Rotator cuff injury	“Lower only until upper arms are parallel to floor”

Pose	Error	Risk	Correction Cue
Chaturanga	Elbows flaring outward	Shoulder impingement	“Hug elbows close to ribs”
Chair	Knees past toes with forward lean	Patellofemoral compression	“Sit hips back; weight in heels”
Tree	Foot placed on the knee	Lateral knee force	“Foot above or below the knee—never on it”
Pigeon	Front shin forced parallel, exceeding hip ROM	Lateral meniscus stress	“Shin angle depends on hip mobility; flex front foot”
Cobra	Arms fully straightened, lumbar compression	Facet joint impingement	“Keep elbows bent; lift from back muscles, not hands”
Bridge	Knees splaying outward	Asymmetric sacral load	“Keep knees hip-width; block between thighs”
Plank	Hips sagging below shoulder–ankle line	Lumbar hyperextension	“Engage core; one line from crown to heels”
Camel	Head dropping back without cervical control	Cervical facet compression	“Keep chin slightly tucked; head is last to go back”
Forward Fold	Rounded lower back with straight legs	Lumbar disc loading	“Bend knees generously; flat back more important than straight legs”

Detectable vs. Undetectable Errors

Not all errors in table 6 are reliably detectable from a single monocular view. Knee valgus in Warrior II (frontal plane) is well-captured by a front-facing camera, but lumbar rounding in Forward Fold (sagittal plane) requires a side view. The system should guide the user to an appropriate camera angle per pose category.

12 Inverse Dynamics from Keypoints

With keypoint positions alone, the system can estimate internal joint loads using inverse dynamics—computing the forces and torques required to maintain a static pose under gravity.

12.1 Segmental Body Model

We adopt de Leva’s [11] anthropometric regression tables, which map height and mass to segment inertia parameters for 16 body segments. Given a user’s height H and mass M (entered once during onboarding), each segment i has:

- Mass fraction: $m_i = \alpha_i M$
- Centre of mass location: $\mathbf{r}_i^{CoM}$, interpolated as a fraction λ_i along the proximal–distal axis
- Moment of inertia: $I_i = \beta_i m_i L_i^2$

Table 7. De Leva segment mass fractions (male/female) for key segments [11].

Segment	Male α_i	Female α_i
Head + neck	0.0694	0.0668
Trunk	0.4346	0.4257
Upper arm	0.0271	0.0255
Forearm	0.0162	0.0138
Thigh	0.1416	0.1478
Shank	0.0433	0.0481
Foot	0.0137	0.0129

12.2 Joint Torque Estimation

For a static pose (or slow quasi-static yoga movements), Newton–Euler inverse dynamics simplifies to a torque-balance at each joint. Working from distal to proximal, the net joint torque at joint j is:

$$\boldsymbol{\tau}_j = \sum_{k \in \text{distal}(j)} m_k g \cdot (\mathbf{r}_k^{\text{CoM}} - \mathbf{r}_j) \times \hat{\mathbf{e}}_z$$

where $g = 9.81 \text{ m/s}^2$ and $\hat{\mathbf{e}}_z$ is the vertical unit vector. This gives a gravitational torque demand at each joint, indicating how much muscular effort is required to hold the pose.

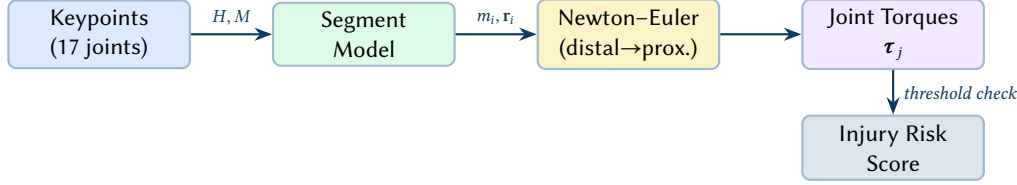


Figure 8. Inverse dynamics pipeline: from keypoints to joint torques and injury risk assessment.

13 Centre of Mass and Balance Analysis

Balance is fundamental to yoga. The system estimates the whole-body centre of mass (CoM) and compares it against the base of support (BoS) to quantify stability.

13.1 CoM Estimation

Using the segmental model from section 12, the whole-body CoM is:

$$\mathbf{r}_{\text{CoM}} = \frac{\sum_{i=1}^N m_i \mathbf{r}_i^{\text{CoM}}}{\sum_{i=1}^N m_i}$$

where segment centroids $\mathbf{r}_i^{\text{CoM}}$ are estimated from keypoint pairs (e.g., the thigh centroid lies 43% proximal along the hip–knee segment for males [11]).

13.2 Base of Support

The BoS is the convex hull of ground contact points. For a 17-keypoint COCO skeleton, ground contacts are estimated from ankle positions plus an estimated foot rectangle:

$$A_{\text{BoS}} = \text{Area}(\text{ConvexHull}(\{\mathbf{p}_i : i \in \text{ground}\}))$$



Figure 9. Base of support comparison: wide stance (Warrior II) vs. single-leg balance (Tree Pose). CoM projection relative to BoS determines stability margin.

13.3 Balance Metrics

Three metrics quantify postural stability during a held pose [12]:

1. **Stability Index:** the minimum distance from CoM projection to the BoS boundary, normalised by BoS diameter:

$$\text{SI} = \frac{\min_{\mathbf{p} \in \partial \text{BoS}} \|\mathbf{p}_{\text{CoM}}^{\text{proj}} - \mathbf{p}\|}{d_{\text{BoS}}}$$

2. **Sway Path Length:** total distance travelled by the CoM projection over T seconds of hold:

$$\text{SPL} = \sum_{t=1}^T \|\mathbf{p}_{\text{CoM}}^{\text{proj}}(t) - \mathbf{p}_{\text{CoM}}^{\text{proj}}(t-1)\|$$

3. **Sway Area:** the 95% confidence ellipse area of CoM displacement, capturing the variance of postural control.

Insight:
Joint torques estimated from keypoints are approximate (2D projection, no ground reaction forces) but sufficient for relative comparisons: “your left knee bears 40% more torque than your right in Warrior II.”

Insight:
Sway path length during a 30-second hold is a sensitive measure of balance improvement over time. A 20% reduction in SPL over 8 weeks is clinically meaningful [12].

14 Injury Risk Modelling

The system combines joint angles, torques, and postural data into a composite injury risk score.

14.1 Biomechanical Thresholds

Table 8. Injury risk thresholds from the biomechanics literature [13, 14].

Risk Factor	Condition	Threshold	Injury Mechanism
Knee valgus	Dynamic valgus angle	$> 12^\circ$	ACL/MCL strain
Lumbar flexion	Flexion under rotation	$> 45^\circ$	Disc herniation
Cervical extension	Extension under load	$> 50^\circ$	Facet compression
Knee flexion (lunge)	Knee past toes	$< 70^\circ$	Patellofemoral stress
Shoulder elevation	Abduction without rotation	$> 120^\circ$	Impingement

14.2 Composite Injury Risk Score

The per-frame injury risk is a weighted sum of normalised risk factors:

$$R_{injury} = \sum_{k=1}^K w_k \cdot \max\left(0, \frac{\theta_k - \theta_k^{safe}}{\theta_k^{danger} - \theta_k^{safe}}\right)$$

where θ_k^{safe} is the threshold below which no risk is flagged, θ_k^{danger} is the hard limit, and w_k weights each factor by severity.

Risk Score Response Curve

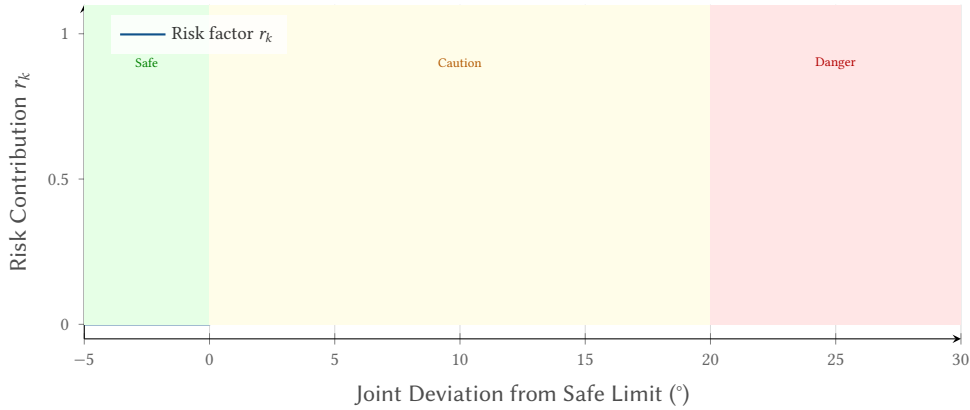


Figure 10. Risk response curve: zero below the safe threshold, linear ramp through the caution zone, saturated at the danger limit.

: Risk Alert Priority

When $R_{injury} > 0.7$, the system should issue an immediate verbal warning. Between 0.3 and 0.7, a gentle visual cue (amber joint marker). Below 0.3, no risk alert is shown.

15 Movement Quality Metrics

Movement smoothness is a hallmark of skilled motor performance. The system quantifies this using metrics from the motor control literature.

15.1 SPARC: Spectral Arc Length

The SPARC metric [15] measures smoothness via the arc length of the normalised Fourier magnitude spectrum of the velocity profile:

$$SPARC = - \int_0^{\omega_c} \sqrt{\left(\frac{1}{\omega_c}\right)^2 + \left(\frac{d\hat{V}(\omega)}{d\omega}\right)^2} d\omega$$

where $\hat{V}(\omega)$ is the normalised Fourier magnitude of the movement speed and ω_c is a cutoff frequency (typically 20 Hz). More negative values indicate less smooth movement.

15.2 Jerk-Based Metrics

The normalised jerk cost [16] penalises the rate of change of acceleration:

$$C_{jerk} = \sqrt{\frac{T^5}{2L^2} \int_0^T \|\ddot{\mathbf{x}}(t)\|^2 dt}$$

where T is the movement duration, L is the path length, and $\ddot{\mathbf{x}}(t)$ is the jerk vector. Lower values indicate smoother movement.

15.3 Hold Stability Index

For static holds, movement quality is captured by the *hold stability index*:

$$HSI = 1 - \frac{\sigma_\theta}{\bar{\theta}}$$

where $\bar{\theta}$ is the mean joint angle during the hold and σ_θ is its standard deviation. An HSI close to 1 indicates a rock-steady hold.

Insight: SPARC is dimensionless, robust to movement amplitude and duration, and monotonically decreasing with the addition of sub-movements—making it ideal for tracking transition quality over practice sessions [15].

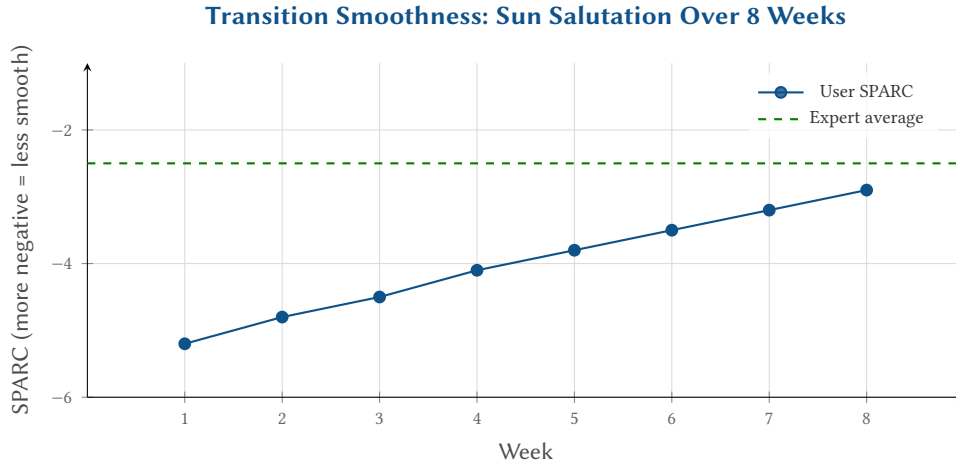


Figure 11. SPARC smoothness improving over weeks as the practitioner develops more fluid transitions.

16 Fatigue and Breath Detection

16.1 Fatigue Detection from Keypoint Drift

As muscles fatigue during a held pose, the practitioner’s posture degrades predictably. The system detects fatigue via three signals:

1. **Angle drift:** a sustained drift in a joint angle $> 1^\circ/\text{min}$ indicates progressive loss of control.
2. **Variance ratio:** compare the variance of joint angles in the first and second halves of a hold. A ratio > 1.5 signals increasing instability.
3. **Tremor detection:** wavelet analysis of keypoint trajectories in the 6–12 Hz band detects muscular tremor preceding failure.

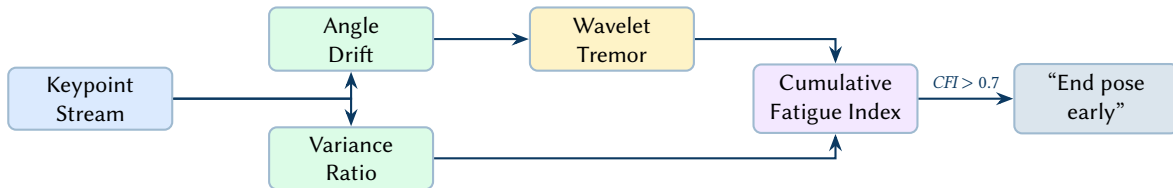


Figure 12. Fatigue detection pipeline: three complementary signals are fused into a cumulative fatigue index (CFI).

The cumulative fatigue index combines these signals:

$$CFI(t) = \alpha \cdot d_{drift}(t) + \beta \cdot r_{var}(t) + \gamma \cdot E_{tremor}(t)$$

: Fatigue-Aware Hold Duration

When CFI exceeds 0.7, the system should recommend ending the pose early, regardless of the target hold time. This overrides timer-based hold recommendations and is a key safety differentiator.

16.2 Breathing Estimation from Shoulder Oscillation

Breathing causes periodic vertical displacement of the shoulder keypoints. By bandpass filtering the shoulder y -coordinate trajectory (0.03–0.5 Hz, corresponding to 2–30 breaths per minute), the system can estimate breathing rate without any wearable:

$$f_{breath} = \frac{1}{T} \cdot \text{count}(\text{peaks}(\text{BPF}(y_{\text{shoulder}}(t))))$$

This enables breath-synchronised cueing even without explicit breath input from the user.

17 Range of Motion Assessment

The system can compare observed joint angles against clinical norms to provide a ROM assessment.

Table 9. Clinical ROM norms vs. yoga demands for key joints [17].

Joint	Movement	Clinical Norm	Yoga Demand
Hip	Flexion	120°	150° (fwd fold)
Hip	Extension	30°	45° (Warrior I back leg)
Hip	Ext. rotation	45°	90° (Lotus)
Shoulder	Flexion	180°	180° (overhead)
Shoulder	Extension	60°	90° (Camel arms)
Knee	Flexion	135°	160° (Hero pose)
Spine	Flexion	60°	90° (full fwd fold)
Spine	Extension	25°	45° (Wheel)

18 Pose Embedding Spaces

Effective pose comparison requires mapping raw keypoints into a compact vector space where geometric proximity reflects semantic similarity.

18.1 Normalised Joint Angle Embeddings

The simplest embedding represents a pose as a vector of joint angles. Given three keypoints forming a joint (parent, vertex, child):

$$\theta_j = \arccos\left(\frac{(\mathbf{p}_{\text{parent}} - \mathbf{p}_{\text{joint}}) \cdot (\mathbf{p}_{\text{child}} - \mathbf{p}_{\text{joint}})}{\|\mathbf{p}_{\text{parent}} - \mathbf{p}_{\text{joint}}\| \cdot \|\mathbf{p}_{\text{child}} - \mathbf{p}_{\text{joint}}\|}\right)$$

For J joints, this yields $\mathbf{e} \in \mathbb{R}^J$, fully invariant to translation, scale, and global rotation. The embedding is augmented with relative bone orientations and dihedral angles for 3D capture, typically yielding $\mathbf{e} \in \mathbb{R}^{50-80}$.

18.2 Graph Neural Network Embeddings

The skeleton is naturally a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with joints as vertices and bones as edges. Graph convolutional networks [18] learn topology-aware embeddings:

$$\mathbf{H}^{(l+1)} = \sigma(\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2} \mathbf{H}^{(l)} \mathbf{W}^{(l)})$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the augmented adjacency matrix. The final node features are aggregated via pooling to produce a fixed-dimensional embedding $\mathbf{e}_{GCN} \in \mathbb{R}^{128-512}$.

18.3 Contrastive Learning for Pose Similarity

Supervised contrastive learning [19] directly optimises embedding similarity to reflect pose class membership:

$$\mathcal{L}_{\text{SupCon}} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_p)/\tau)}{\sum_{a \in A(i)} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_a)/\tau)}$$

Insight:
Yoga routinely demands ROM beyond clinical norms. Framing this data as “your current hip flexion is 110° of a possible 120° clinical norm” gives the practitioner context without encouraging unsafe pushing beyond their anatomy.

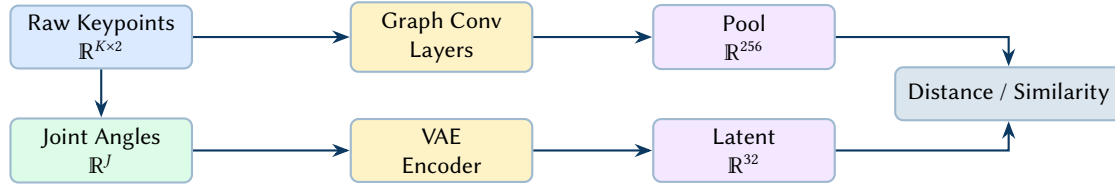


Figure 13. Two embedding pathways: GCN on raw keypoints (learned, high-dimensional) and VAE on joint angles (generative, compact).

where $P(i)$ are same-class positives and τ is a temperature parameter. This produces embeddings where poses of the same type cluster tightly while different poses are well-separated.

Design Decision 4: Embedding Strategy

For real-time classification, use a contrastive GCN embedding (256-d, cosine similarity). For generative tasks (interpolation, sequence synthesis), use a VAE latent space (32-d). Both can share the same GCN backbone with branching heads.

19 Bilateral Asymmetry Quantification

Detecting and tracking left–right imbalances is clinically valuable.

19.1 Symmetry Index

For any bilateral measurement X_L, X_R :

$$SI = \frac{2(X_R - X_L)}{X_R + X_L} \times 100\%$$

This ranges from -200% to $+200\%$, with 0% indicating perfect symmetry. For joint angles, segment lengths, or ROM values.

19.2 Procrustes Asymmetry

A more rigorous approach uses Procrustes analysis [10]: reflect the right-side landmarks across the sagittal plane, optimally align via SVD, and compute the residual Procrustes distance $d_P = \|\mathbf{L} - \mathbf{R}\hat{\mathbf{R}}\|_F$.

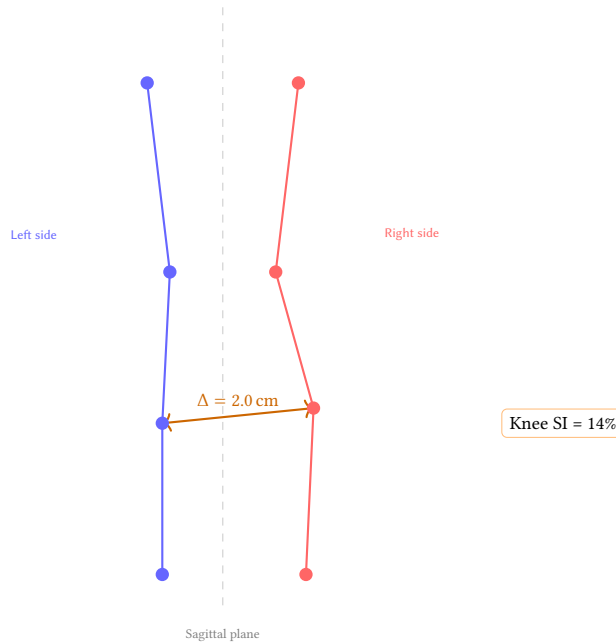


Figure 14. Bilateral asymmetry visualisation: left vs. right side keypoints overlaid with asymmetry measurement.

Insight:
An increasing asymmetry trend over sessions is more clinically relevant than a stable one, even if the absolute value is lower.

Table 10. Asymmetry interpretation guidelines.

SI Range	Status	Interpretation
<10%	Normal	Within biological variation
10–15%	Monitor	Handedness effects; track trends
>15%	Attention	Consistent favouring; suggest corrective focus
>25%	Flag	Possible injury compensation; recommend assessment

20 Pose Difficulty Estimation

A computational difficulty score enables adaptive sequencing and progressive programming.

20.1 Biomechanical Difficulty Factors

The composite difficulty score combines six biomechanical factors:

$$D_{total} = w_1 D_{balance} + w_2 D_{height} + w_3 D_{ROM} + w_4 D_{inversion} + w_5 D_{support} + w_6 D_{strength}$$

where:

- $D_{balance} = A_{ref}/A_{BoS}$ – inversely proportional to base of support area.
- $D_{height} = h_{CoM}/h_{standing}$ – CoM height relative to standing.
- $D_{ROM} = \frac{1}{j} \sum_j \theta_j / \theta_j^{max}$ – mean joint ROM utilisation.
- $D_{inversion} = \alpha_{torso} / \pi$ – torso angle from vertical.
- $D_{support} = (4 - n_{support})/3$ – fewer support limbs = harder.
- $D_{strength}$ – estimated from joint torque demands (section 12).

Estimated Difficulty Scores for Common Poses

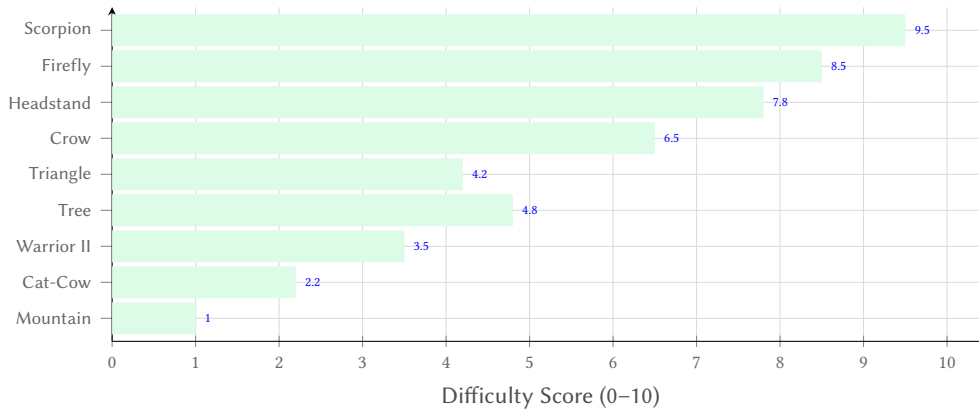


Figure 15. Estimated difficulty scores for a range of poses, computed from the composite biomechanical model.

21 Temporal Sequence Analysis

Beyond single-frame pose comparison, the system analyses temporal sequences for flow recognition and quality assessment.

21.1 Hidden Markov Models for Flow Structure

An HMM models yoga flows as emissions from a Markov chain:

- **States:** canonical poses and transition phases.
- **Transition matrix A:** encodes expected pose-to-pose sequencing.
- **Emission model:** Gaussian mixture over the pose embedding space.

The Viterbi algorithm finds the most likely state sequence, enabling automatic temporal segmentation of a continuous practice recording into discrete pose labels.

21.2 Spatio-Temporal Graph Convolutions

For skeleton-based action recognition, ST-GCN [18] and its successors (CTR-GCN [20]) process a tensor $\mathbf{X} \in \mathbb{R}^{C \times T \times V}$ (channels, frames, vertices) through interleaved spatial graph convolutions and temporal convolutions:

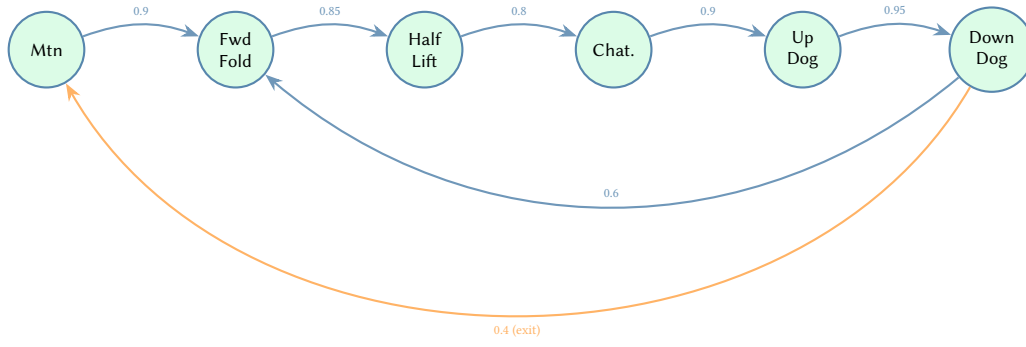


Figure 16. HMM state diagram for Sun Salutation A. Transition probabilities encode the expected flow structure. Low-probability transitions signal sequencing errors.

1. **Spatial:** graph convolution within each frame models joint-joint relationships.
2. **Temporal:** 1D convolution along the time axis models how each joint moves over time.

Design Decision 5: Yoga-Specific Adaptations for ST-GCN

Standard action recognition pipelines downsample to 2–5 s clips. Yoga flows require: (1) longer sequence windows (30–60 s for a Sun Salutation), (2) finer temporal resolution (slow transitions), (3) temporal segmentation (classify *and* segment simultaneously via MS-TCN [21]), and (4) quality regression heads alongside classification.

21.3 Transition Quality via Pose Manifold Geodesics

Given a learned pose latent space (VAE [22]), the expected transition between two poses follows the geodesic on the pose manifold. The observed transition is compared against this:

$$\text{PathDeviation}(t) = d(\mathbf{x}_{\text{observed}}(t), \gamma^*(t))$$

where γ^* is the expected geodesic trajectory. Large deviations indicate compensatory patterns, balance corrections, or incorrect sequencing. Motion diffusion models [23] can generate reference trajectories for transition quality scoring.

22 Anomaly Detection in Movement

The system should flag unusual or potentially unsafe movements that don’t match any known pose pattern.

22.1 Reconstruction-Based Detection

A variational autoencoder trained on “normal” pose sequences provides an anomaly score via reconstruction error:

$$\text{AnomalyScore}(\mathbf{x}) = -\text{ELBO}(\mathbf{x}) = -\mathbb{E}_{q(\mathbf{z}|\mathbf{x})}[\log p(\mathbf{x}|\mathbf{z})] + D_{KL}(q(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))$$

High scores indicate either an unusual pose (poor reconstruction) or a pose in a low-density latent region.

22.2 Practical Anomaly Pipeline

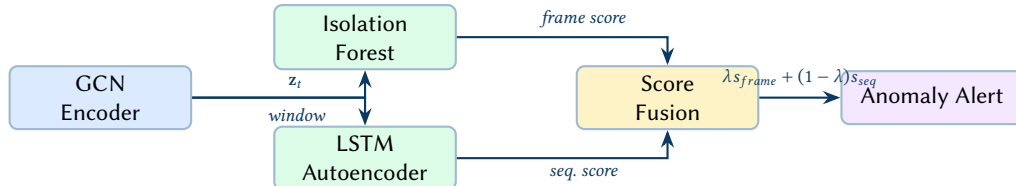


Figure 17. Two-level anomaly detection: frame-level (Isolation Forest [24]) catches unusual static poses; sequence-level (LSTM autoencoder) catches unusual transitions.

23 Body Proportion Normalisation

Fair pose comparison across different body types requires normalisation that separates pose from body shape.

23.1 Skeleton Retargeting

The most physically meaningful approach copies local joint rotations from the observed skeleton to a standard skeleton:

$$\mathbf{q}_j^{\text{standard}} = \mathbf{q}_j^{\text{observed}} \quad \forall j$$

where $\mathbf{q}_j$ is the local rotation of joint j relative to its parent. This maps all bodies to a canonical shape while perfectly preserving pose, yielding a representation $\mathbf{e} = [\mathbf{q}_1, \dots, \mathbf{q}_J] \in (SO(3))^J$ that is completely body-shape invariant.

23.2 Generalised Procrustes Normalisation

An alternative uses Generalised Procrustes Analysis [10]: collect the same pose from multiple body types, iteratively align all instances to their mean shape (translate, scale, rotate), and use the converged mean as the normalisation target. New poses are normalised by Procrustes-aligning to this mean.

: Normalisation Strategy

Use skeleton retargeting (joint rotations onto a standard skeleton) for all comparison and classification tasks. Reserve Procrustes normalisation for raw keypoint visualisation (e.g., before/after overlays) where rotations are not extracted.

24 Kinetic Chain Compensation Detection

Compensation patterns are distinct from simple alignment errors: they involve *multi-segment coordination* where one body region “cheats” to overcome a restriction elsewhere.



Figure 18. Compensation detection: multi-joint conditional rules identify cheating patterns and trace them to the underlying restriction.

Example compensation chains:

- **Forward fold:** hip lateral shift $> 5^\circ$ while torso lean $<$ expected $\Rightarrow$ hip flexor tightness compensating for hamstring restriction.
- **Overhead reach:** lumbar hyperextension $> 15^\circ$ during shoulder flexion $\Rightarrow$ limited thoracic extension or tight lats compensated by the lower back.
- **Warrior I back leg:** pelvis rotating open $> 20^\circ \Rightarrow$ tight hip flexors preventing square alignment.

: Compensation Rules as TOML

Define compensation rules in a human-readable TOML format (IF-THEN-BECAUSE) loadable without recompilation. This allows physiotherapists and senior teachers to author new rules as the knowledge base grows.

KNOWLEDGE LAYER

The intelligence layer quantifies *what the body is doing*; the knowledge layer supplies *what the body should be doing* and *why*. Here we encode the domain expertise that distinguishes a yoga-aware system from a generic motion tracker: anatomy, sequencing theory, pose genealogy, philosophy, and contextual factors like circadian rhythm.

25 Myofascial Meridians and Muscle Engagement

Understanding *which tissues a pose affects* transforms the app from a shape-matcher into a meaningful practice companion.

25.1 Major Fascial Lines

25.2 Pose-to-Fascial-Line Mapping

Insight:
Thomas Myers' *Anatomy Trains* model [25] provides a tractable framework for mapping poses to continuous fascial chains, explaining why effects propagate beyond the apparent stretch site.

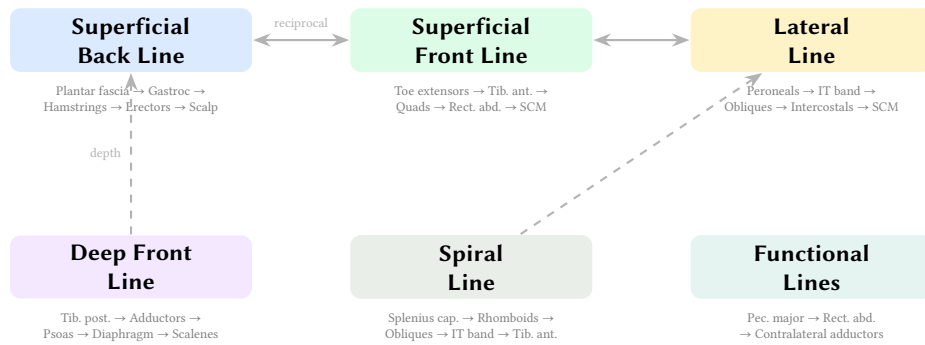


Figure 19. Major myofascial meridians and their muscle chains [25].

Table 11. Primary fascial line engagement for 10 representative poses.

Pose	Primary Line	Action	Key Muscles
Downward Dog	SBL	Stretch	Hamstrings, calves, erectors
Upward Dog/Cobra	SFL	Stretch	Rect. abdominis, hip flexors
Triangle	Lateral	Stretch	IT band, obliques, intercostals
Revolved Triangle	Spiral	Engage	Obliques, serratus anterior
Pigeon	DFL	Stretch	Psoas, iliacus (back leg)
Warrior I	SFL	Stretch	Hip flexors (back leg)
Seated Fwd Fold	SBL	Stretch	Full posterior chain
Side Plank	Lateral	Strengthen	Peroneals, obliques, intercostals
Wheel	SFL	Stretch	Full anterior chain
Eagle	Deep Back Arm	Stretch	Rhomboids, rotator cuff

26 Counter-Poses and Sequencing

26.1 Counter-Pose Theory

Counter-poses (*pratikriyasana*) serve four biomechanical purposes:

1. **Reciprocal loading:** promotes symmetric disc hydration by alternating flexion and extension.
2. **Neurological reset:** activates antagonists via reciprocal inhibition, reducing residual tone.
3. **Fascial creep recovery:** reloads tissue at its new length after sustained stretch.
4. **Spinal disc mechanics:** alternates nucleus pulposus migration direction.

: Counter-Pose Intensity

A counter-pose should be 50–70% the intensity of the primary pose and held for a shorter duration. The goal is restoration to neutral, not maximal stretch in the opposite direction.

Table 12. Common pose–counterpose pairs with biomechanical rationale.

Primary Pose	Counter-Pose	Rationale
Wheel	Knees to chest (Apanasana)	Decompress lumbar facets
Seated Fwd Fold	Gentle Cobra/Sphinx	Reverse posterior disc migration
Shoulderstand	Fish Pose (Matsyasana)	Reverse cervical flexion
Headstand	Child’s Pose (Balasana)	Decompress cervical spine
Camel	Rabbit/Child’s Pose	Reverse thoracic extension
Pigeon	Tabletop/Downward Dog	Return hip to neutral rotation
Warrior series	Wide-legged Fwd Fold	Release isometric leg fatigue
Arm balances	Wrist circles/reverse prayer	Address wrist compression

26.2 Sequencing Principles

Key sequencing constraints that the system must enforce:

- **No cold hamstring loading:** deep forward folds should not appear in the first 5 minutes.

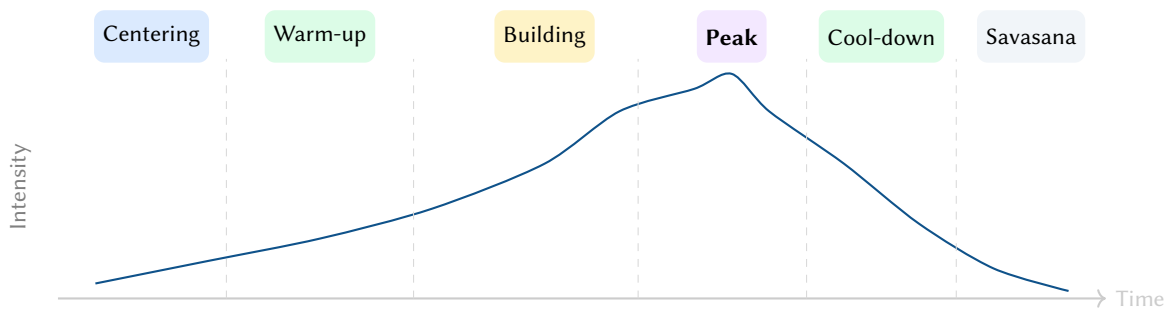


Figure 20. The universal arc of a yoga class: intensity follows a wave pattern peaking before the cool-down.

- **No deep backbend → deep forward fold without passing through neutral:** rapid spinal reversal at high amplitude risks disc injury.
- **No deep twists immediately after backbends:** facet joints are already compressed; adding rotation under compression increases injury risk.
- **Complete both sides** of asymmetric poses before counter-posing.
- **Progressive preparation:** the peak pose's required joint actions must be warmed up by prior poses.

Contraindicated Transitions

These transitions should be **blocked** by the sequence builder: yin-style long holds (>3 min) followed immediately by dynamic vinyasa targeting the same tissue group. Fascia undergoing prolonged creep is temporarily less stiff and more vulnerable to high-force loading.

26.3 Hold Time Recommendations

Table 13. Recommended hold durations by style and pose type.

Style	Duration	Tissue Target	Rationale
Vinyasa/Power	15–30 s (3–5 breaths)	Muscle strength, cardio	Breath–movement sync
Hatha	30–60 s (5–10 breaths)	Proprioception, alignment	Time for refinement
Yin	3–5 min	Fascia, ligaments	Viscoelastic creep
Restorative	5–20 min	Parasympathetic NS	Full support, no effort
Ashtanga	25–30 s (5 breaths)	Balanced	Standardised system

27 Yoga Pose Datasets

Table 14. Available yoga pose datasets and taxonomies.

Dataset	Classes	Size	Annotation Type
Yoga-82 [26]	82	~28,400 images	Classification labels
Kaggle (5-class)	5	~1,500 images	Classification labels
Kaggle (107-class)	107	Variable	Classification labels
Light on Yoga [27]	~200	Textual	Difficulty ratings, descriptions
COCO [28]	N/A	200k+ images	17 keypoints

Insight:
If form degrades during a hold, the system should recommend ending early regardless of target duration—detecting form degradation over time is a key differentiator over a static timer.

Bootstrapping the Reference Database

No single dataset provides both keypoint annotations and yoga pose labels for the full range of common poses. The recommended approach: use a COCO-trained pose estimator (RTMPose) to extract keypoints from Yoga-82 images, then compute reference angle distributions per pose class. This yields both a pose classifier training set and per-pose angle reference ranges.

28 Pose Genealogy and Variation Tree

Each yoga pose exists within a family of variations—progressions, regressions, and lateral variations that share biomechanical DNA.

The system uses the variation tree for:

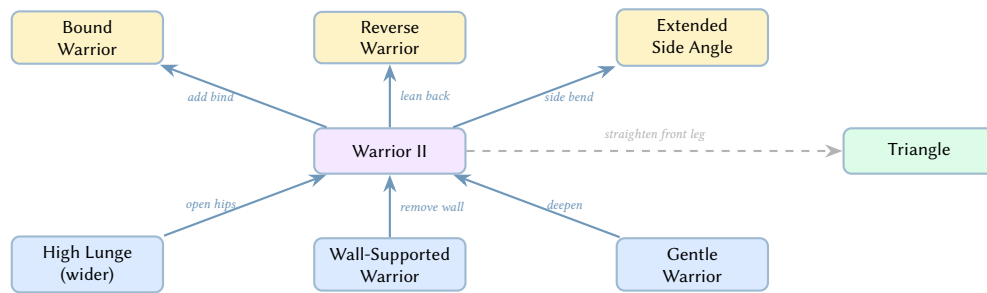


Figure 21. Pose variation tree for Warrior II: progressions (harder), regressions (easier), and lateral variations (related poses).

- **Automatic regression:** if the user cannot maintain form in a pose, suggest the nearest regression in the tree.
- **Progression unlocks:** when all correction rules pass consistently for a pose, suggest the next progression.
- **Exploration prompts:** “you’ve mastered Warrior II. Have you tried Extended Side Angle?”
- **Sequence generation:** traverse the tree to build sequences with natural biomechanical progression.

29 Yoga Philosophy and Sanskrit Integration

Beyond the physical, the app can integrate the wider yoga tradition:

- **Sanskrit name pronunciation:** audio playback of correct Sanskrit pose names with phonetic transcription.
- **Etymological context:** explain the meaning behind pose names (e.g., Virabhadrasana = Warrior, named after a fierce deity; Savasana = Corpse Pose, representing surrender).
- **Chakra mapping:** overlay the seven major chakra locations on the body, highlighting which energy centres each pose is traditionally associated with.
- **Pranayama guidance:** structured breathing exercises (Ujjayi, Nadi Shodhana, Kapalabhati) with visual breath pace guides, tracked via the shoulder-oscillation breath detection (section 16).
- **Drishti cues:** indicate the recommended gaze point (drishti) for each pose, which aids balance and focus.
- **Mudra recognition:** if hand landmarks are available (BlazePose 33-point), detect common hand mudras (Gyan, Chin, Anjali) during meditation and pranayama.

30 Biomechanical Digital Library

A comprehensive, searchable digital library of pose biomechanics that grows with the user base.

1. **Per-pose biomechanical profile:** for each of 200+ poses, store the distribution of joint angles, torques, fascial loads, difficulty scores, and common errors observed across the user population (privacy-preserving, cf. section 68).
2. **Comparative anatomy viewer:** “how does this pose differ for someone with long femurs vs. short femurs?” Render the standard skeleton with adjustable body proportions, showing how the ideal alignment changes with anatomy.
3. **Contraindication database:** structured, evidence-based contraindications per pose per condition (pregnancy trimester, osteoporosis, hypertension, glaucoma, etc.), sourced from clinical yoga therapy literature.
4. **Literature citations:** each correction rule and safety threshold links to its source in the biomechanics or yoga therapy literature, building trust and enabling verification.

31 Circadian and Ultradian Rhythm Integration

The body’s flexibility, strength, and injury risk vary predictably throughout the day.

The system adjusts session parameters based on practice time:

- **Early morning:** extended warm-up, gentler initial ROM targets, avoid cold-tissue deep stretches.
- **Afternoon:** optimal for challenging poses, deeper stretches, strength-focused sequences.
- **Evening:** favour calming sequences, avoid stimulating backbends close to bedtime, incorporate parasympathetic-activating poses.
- **Ultradian rhythm:** every 90–120 minutes the body cycles through higher and lower alertness. If wearable data shows a low-alertness trough, suggest a shorter, gentler session.

Circadian Variation in Key Practice Parameters

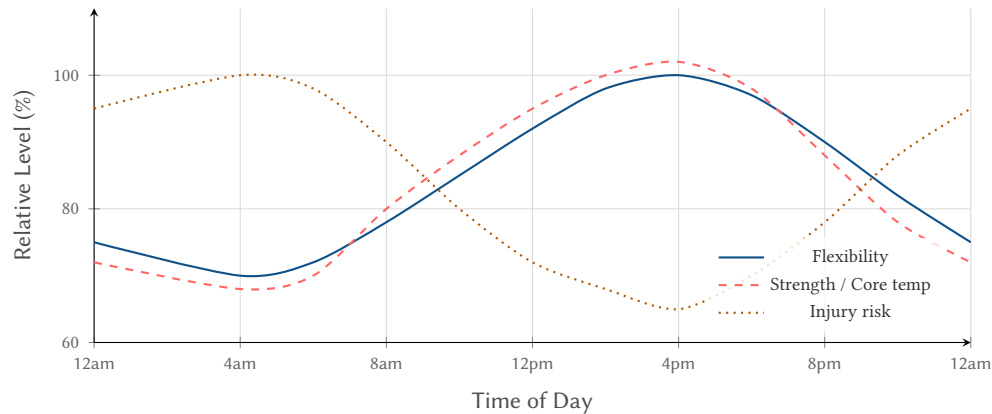


Figure 22. Circadian variation: flexibility and strength peak in the afternoon; injury risk is highest in the early morning when tissues are cold and stiff.

32 Thermal and Muscle Activation Estimation

The system can estimate which muscles are actively engaging vs. passively stretching, enabling a “muscle activation map” overlay.

1. **Isometric estimation:** from joint angles and estimated torques (section 12), infer which muscles must be contracting to maintain the pose. Muscles acting against gravity at a joint are classified as “active”; muscles on the opposite side as “stretched.”
2. **Synergy groups:** map individual muscle activations to known muscle synergies (co-contraction patterns) from the motor control literature.
3. **Caloric estimation:** approximate energy expenditure from estimated muscle activation levels and hold durations, using MET (Metabolic Equivalent of Task) values per pose category.
4. **Thermal model:** a simplified bioheat equation predicts local tissue temperature rise from sustained muscle activation, indicating which regions are “warming up” and which remain cold—informing warm-up completeness.

FEEDBACK AND INTERACTION LAYER

Intelligence and knowledge are only valuable if they reach the practitioner at the right moment and in the right modality. This part explores the full spectrum of feedback channels—from augmented-reality anatomy overlays and haptic wearables to soundscape generation and LLM-powered natural-language explanation.

33 Anatomical Augmented Reality Overlay

Beyond the skeleton wireframe, the system can overlay anatomical information directly onto the practitioner’s body in the camera view.

Overlay modes, selectable by the user:

1. **Skeleton only:** wireframe with joint markers (default).
2. **Fascial lines:** colour-coded bands along the active myofascial meridians (fig. 23).
3. **Muscle groups:** highlight stretched (red) vs. contracted (blue) vs. stabilising (green) muscle regions.
4. **Joint angles:** live numeric readouts at each measured joint, colour-coded green/amber/red against the target range.
5. **Force vectors:** arrows showing the direction of gravity load and muscle engagement at key joints.

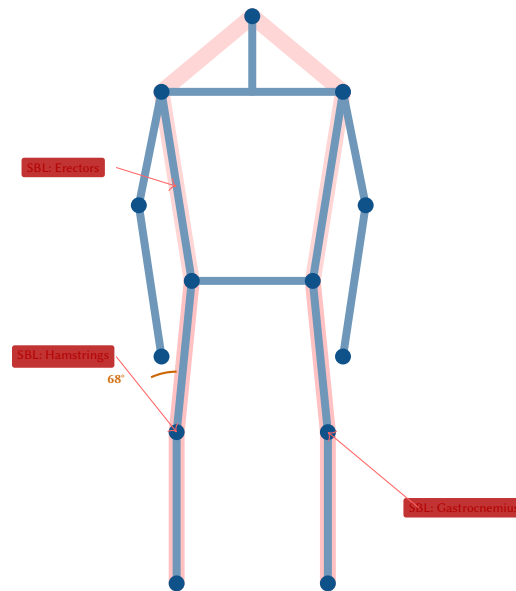


Figure 23. Anatomical AR overlay concept: fascial lines rendered as semi-transparent coloured bands over the skeleton, with muscle group labels and live angle readouts.

34 Breath Synchronisation and Audio Cueing

Breath is fundamental to yoga practice. The system can provide audio and visual breath cueing synchronised to pose transitions and hold phases.

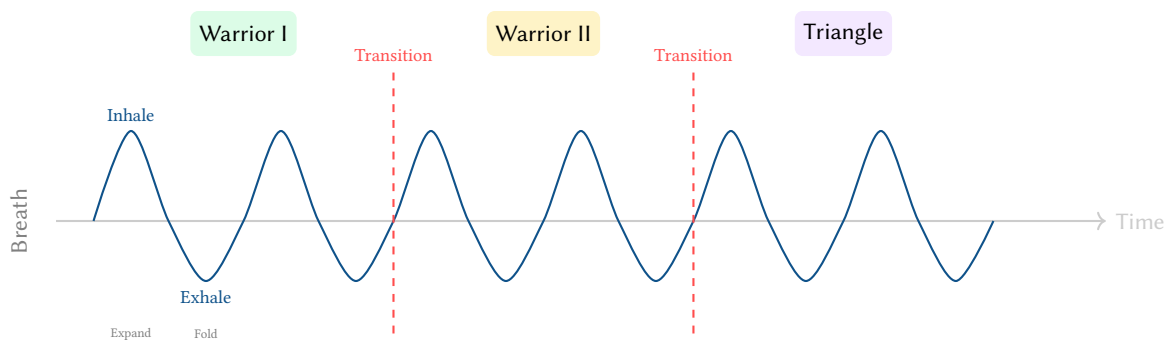


Figure 24. Breath synchronisation: transitions are timed to occur on exhalations (contraction/folding) or inhales (expansion/lifting), following vinyasa convention.

- **Vinyasa cueing:** inhale = expansion, extension, lifting; exhale = contraction, flexion, folding, twisting.
- **Hold-phase breathing:** during static holds, the system provides a gentle pulsing visual or audio metronome at the user's preferred breath rate (default: 4 s inhale, 6 s exhale).
- **AI voice instructor:** text-to-speech for real-time verbal cues ("Inhale, reach your arms overhead... exhale, fold forward") so the practitioner does not need to watch the screen. Can use platform TTS (macOS AVSpeechSynthesizer, Web Speech API) or an offline neural TTS model.

35 Haptic Feedback via Wearables

Visual and audio feedback fail when the user cannot see the screen or is in a noisy environment. Haptic feedback via a smartwatch or vibration bands provides a modality-independent correction channel.

Insight: Voice cueing decouples the user from the screen, which is essential for poses where the screen is not visible (face-down, inversions).

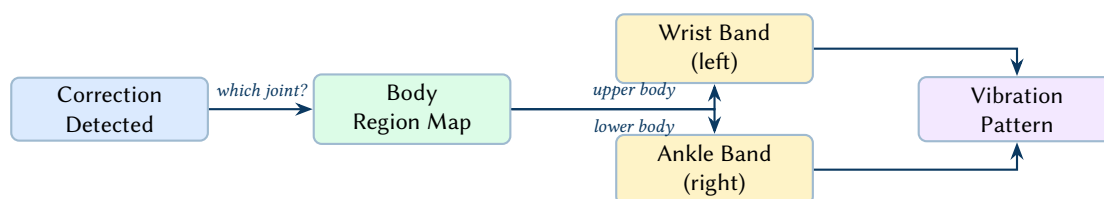


Figure 25. Haptic feedback: corrections are mapped to the nearest vibration band, with directional patterns indicating the required adjustment.

Vibration patterns encode the correction type:

- **Pulse frequency:** faster pulses = more urgent correction.
- **Duration:** continuous buzz = “hold this position”; intermittent = “adjust.”
- **Directional encoding:** two motors per band can encode “move left” vs. “move right” via asymmetric activation.

36 Environmental Soundscape Generation

Procedurally generated ambient audio matched to the session arc, without requiring music licensing.

- **Generative ambient:** use procedural audio synthesis (Rust crates: `rodio` for playback, custom DSP for synthesis) to generate evolving soundscapes—wind, rain, singing bowls, drone tones—synchronised to the session phase.
- **Binaural beats:** optional binaural beat underlays at frequencies associated with different brain states: theta (4–8 Hz) for deep relaxation during Savasana, alpha (8–12 Hz) for focused standing work.
- **Breath-reactive audio:** the soundscape responds to the detected breath rhythm—swelling on inhale, receding on exhale—creating a feedback loop that reinforces breath awareness.
- **Silence as design:** in meditation and Savasana phases, the system can enforce true silence (muting notifications, dimming screen) rather than adding more audio.

37 Digital Twin and 3D Avatar

A full 3D avatar reconstructed from the user’s body proportions (calibrated via an initial standing pose) enables several advanced features.

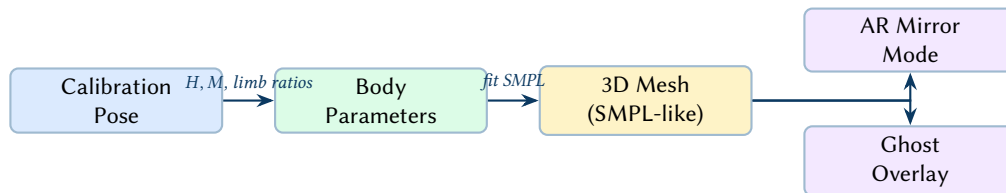
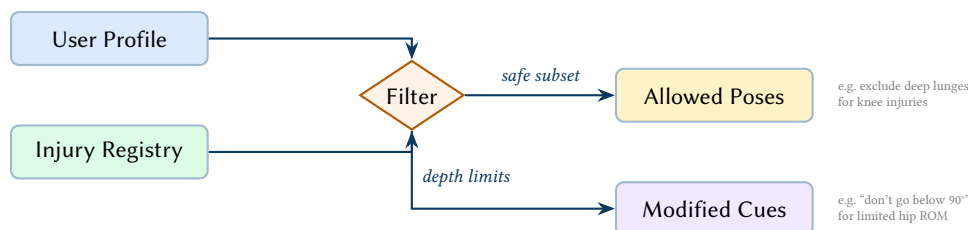


Figure 26. Digital twin pipeline: a single calibration pose parameterises a 3D body model used for AR mirror and ghost overlay.

- **AR Mirror Mode:** the avatar mirrors the user’s movements in real time, rendered from any angle. The user can “orbit” the camera to see their pose from the side or back without a physical mirror.
- **Ghost Overlay:** a semi-transparent “ideal” avatar shows the target pose overlaid on the user’s actual pose, making deviations immediately visible.
- **Time-Lapse Replay:** replay past sessions as a 3D animation, scrubbing through time to see specific moments.
- **Pose Explorer:** browse poses in a 3D catalogue, with the user’s own body proportions, to understand what each pose looks like for *their* body shape before attempting it.

38 Injury-Aware and Personalised Modes



Insight:
Ghost overlay is the single most intuitive feedback mechanism: seeing your body superimposed on the ideal immediately communicates what needs to change, without any verbal instruction.

Figure 27. Injury-aware filtering: the user’s injury profile constrains both the pose set and the depth of individual corrections.

The user marks affected joints or conditions (e.g., “left knee MCL sprain”, “lower back disc herniation”, “pregnancy”). The system:

1. **Excludes** contraindicated poses (e.g., deep twists for disc herniation, inversions for pregnancy).
2. **Modifies** correction thresholds (e.g., “don’t lunge deeper than 110° at the knee”).
3. **Suggests alternatives** (e.g., Half Pigeon → Supine Figure-Four for knee sensitivity).
4. **Flags risk** if the user enters a pose that conflicts with their profile.

Medical Disclaimer

Injury-aware mode provides general guidance based on biomechanical principles. It is not a substitute for medical advice. The system should display a clear disclaimer and recommend consulting a healthcare professional for specific injuries.

39 Dream Sequence Reconstruction

A creative mode where the user describes a desired physical experience in natural language and the system generates a matching sequence.

Table 15. Natural language intent to sequence generation.

User Intent	Generated Focus
“Open my chest and shoulders”	Backbends, chest openers, SFL stretches
“Calm my nervous system”	Forward folds, inversions, long holds, pranayama
“Prepare for a long run”	Hip flexors, hamstrings, ankle mobility, dynamic
“Release tension from sitting all day”	Hip openers, spinal twists, DFL release
“Build toward handstand”	Core, wrist prep, shoulder strength, wall drills
“Something gentle for a hangover”	Restorative, twists, hydrating poses, no inversions

This leverages the LLM integration (section 40) to parse intent, the fascial line database (section 25) to map intent to tissue targets, and the constraint-satisfying sequence generator (section 61) to produce a valid sequence.

40 Offline Knowledge and LLM Integration

While the pose correction engine is rule-based for determinism and speed, a language model can augment the system in non-real-time contexts:

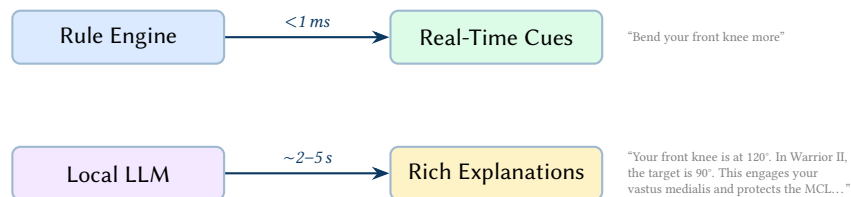


Figure 28. Dual-path feedback: rule engine provides real-time cues; a local LLM generates rich explanations on demand.

- **Post-session summaries:** “Today you practised 12 poses over 25 minutes. Your Warrior II alignment improved. Your hamstring flexibility is 8% better than your 30-day average. Suggested focus for next session: hip openers.”
- **Natural language Q&A:** “Why should I keep my knee over my ankle in Warrior II?” → LLM generates an explanation grounded in the knowledge base.
- **Custom sequence generation:** “Build me a 20-minute hip-opening sequence for an intermediate practitioner” → LLM proposes a sequence, validated against the constraint rules.
- **Offline model:** a small local model (e.g., Phi-3, Llama-3 8B quantised) via `candle` or `llama-cpp-rs` keeps all data on-device.

Design Decision 6: Rule Engine + LLM Hybrid

Real-time corrections **must** use the deterministic rule engine (sub-millisecond, predictable, safe). The LLM is reserved for non-time-critical, explanatory, and generative features. This separation ensures that safety-relevant feedback (“don’t go deeper, your knee is at risk”) is never subject to LLM hallucination.

41 Self-Myofascial Release Prescription

When compensation patterns or ROM deficits persist across 3+ sessions, the system generates a targeted self-myofascial release protocol.

1. Map the ROM deficit to the responsible muscle group(s) using the fascial line database (section 25).
2. Look up trigger point locations for those muscles in a curated anatomical database.

3. Generate a guided foam rolling / ball release protocol with: target body region, tool recommendation (foam roller, lacrosse ball, tennis ball), duration, pressure cues, and a visual body-map showing the release location.
4. Track ROM changes pre- and post-release to validate effectiveness.

LONGITUDINAL INTELLIGENCE

A single session provides a snapshot; sustained practice generates a trajectory. This part develops the time dimension: progress tracking, predictive analytics, periodisation, movement signatures, and the somatic and neuroplastic dimensions of long-term learning.

42 Progress Tracking and Adaptive Practice

Beyond single-session correction, the system can track practitioner progress over weeks and months, building a personalised model of flexibility, strength, and practice patterns.

42.1 Longitudinal Flexibility Tracking

Flexibility Progress: Forward Fold (Hip Angle)

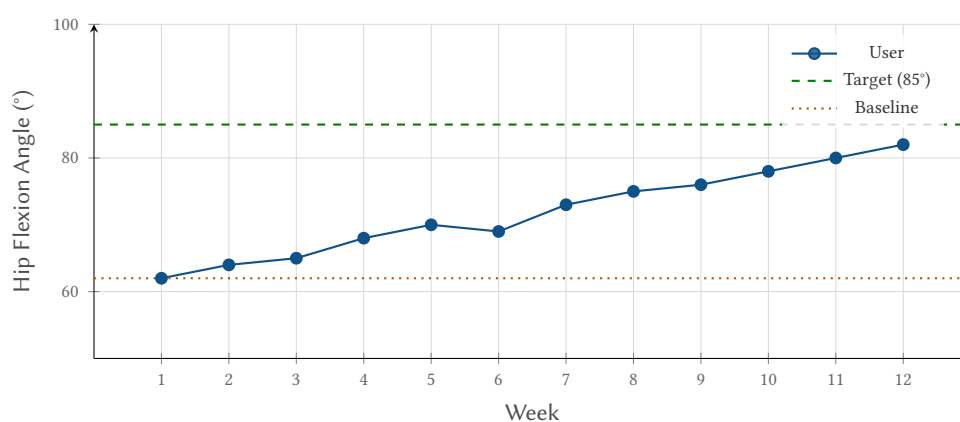


Figure 29. Longitudinal flexibility tracking for a single pose. The system records joint angle ranges per pose per session, revealing measurable progress over weeks.

Each session produces a timestamped record of:

- Per-pose joint angle ranges (min, max, mean during hold).
- Hold duration achieved before form degradation.
- Poses attempted and correction frequency.
- Session duration, sequence composition.

42.2 Adaptive Difficulty Engine

The system monitors which poses consistently trigger corrections and which are held cleanly, building a per-user difficulty profile.

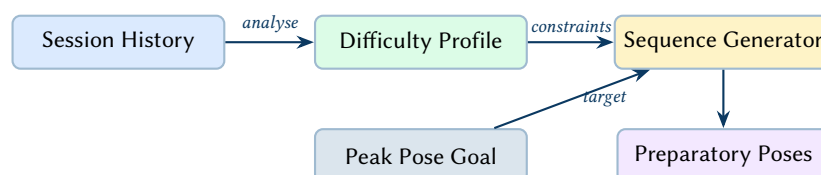


Figure 30. Adaptive difficulty: session history feeds a difficulty profile that constrains the sequence generator.

Insight:
Bridging yoga practice with myofascial release transforms the app from a single-activity tool to a comprehensive movement health platform. The camera can even guide release technique by tracking body positioning over the roller.

Insight:
Tracking angle improvement over time provides objective evidence of progress—something instructors can observe intuitively but rarely quantify.

Design Decision 7: Peak Pose Planner

The user sets a target pose (e.g., Crow Pose, Full Splits). The system identifies prerequisite joint ranges and strength requirements, compares against the user's current profile, and generates a multi-week progressive plan. Each session includes poses that incrementally build the required flexibility and strength, with progress gates ("unlock Crow when wrist extension reaches 75° and core hold exceeds 45 s").

43 Session Recording and Playback

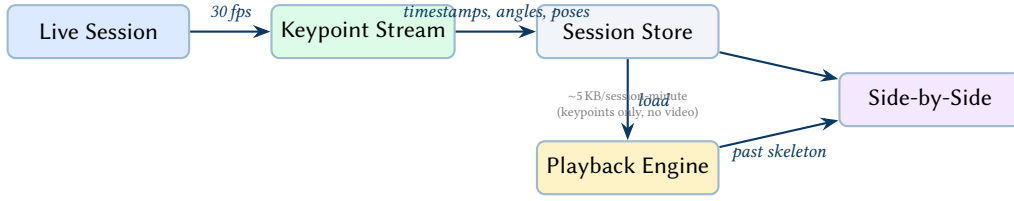


Figure 31. Session recording stores keypoint streams (not video), enabling lightweight playback and before/after comparison.

Key design choice: store **keypoint data**, not video. A 17-keypoint skeleton at 30 fps produces ~5 KB per minute of practice (vs. ~30 MB/min for compressed video). This enables:

- Months of session history in <10 MB.
- Side-by-side skeleton overlay: “you in Week 1” vs. “you today”, aligned by pose via Dynamic Time Warping (DTW) [29].
- Privacy: no video stored on device, only abstract skeleton data.
- Export as a shareable animation or progress report.

44 Longitudinal Practice Intelligence

The richest value of the system emerges from analysing practice patterns over weeks and months.

44.1 Practice Pattern Profiling

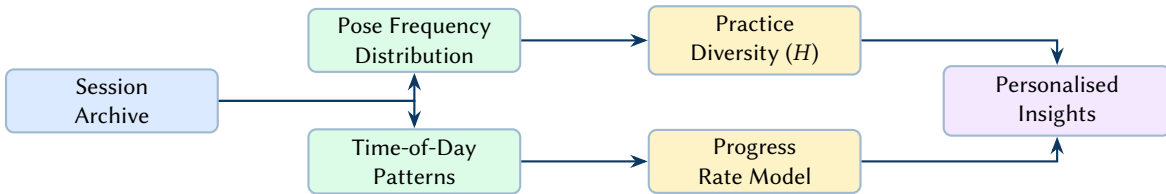


Figure 32. Practice intelligence pipeline: session data is analysed for frequency, diversity, and temporal patterns to generate personalised insights.

Key analytics:

- **Pose preference entropy:** Shannon entropy of the pose frequency distribution measures practice diversity: $H = -\sum_i p_i \log_2 p_i$. Low entropy indicates a narrow practice; the system can suggest under-represented pose categories.
- **Progress rate:** exponential fit to per-pose joint angle improvements, enabling plateau detection (when the rate drops below a threshold).
- **Consistency score:** regularity of practice frequency (e.g., coefficient of variation of inter-session gaps).
- **Time-of-day patterns:** cluster sessions by time to recommend optimal practice windows.

45 Predictive Analytics

45.1 Goal Pose Timeline Prediction

Given a target pose (e.g., full splits) and the user’s current ROM trajectory, the system fits an asymptotic growth model:

$$\theta(t) = \theta_{max} \cdot (1 - e^{-\lambda(t-t_0)}) + \theta_0$$

where θ_0 is the baseline, θ_{max} is the estimated asymptote, and λ is the learning rate. The predicted time-to-goal is:

$$t_{goal} = t_0 - \frac{1}{\lambda} \ln\left(1 - \frac{\theta_{goal} - \theta_0}{\theta_{max}}\right)$$

45.2 Injury Risk Prediction

By tracking compensatory patterns over time (e.g., consistently favouring one side, increasing asymmetry), the system can flag elevated injury risk before symptoms appear:

1. **Compensatory pattern detection:** if asymmetry SI increases > 2% per week for 3+ consecutive weeks.
2. **Overtraining signals:** declining hold stability combined with increasing practice frequency.

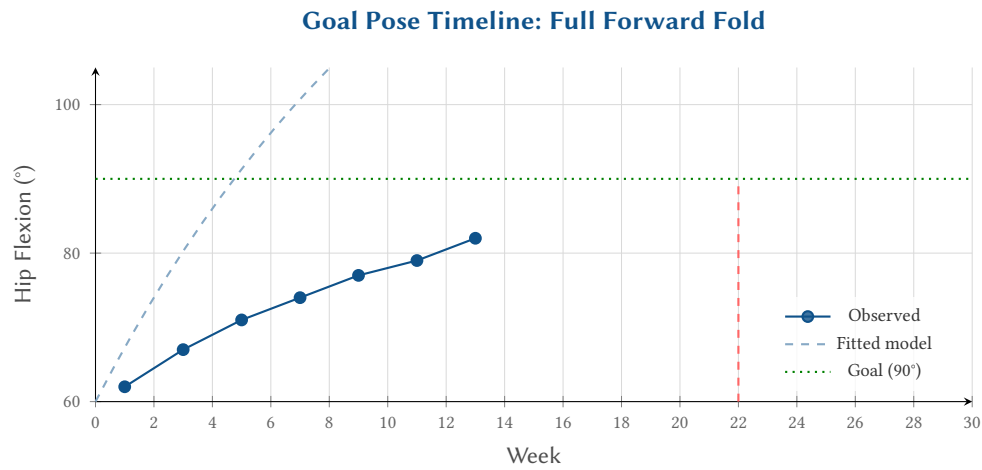


Figure 33. Goal timeline prediction: an asymptotic model fitted to observed ROM data predicts when the target angle will be reached.

3. **Fatigue accumulation:** ACWR (acute:chronic workload ratio) [30] exceeding 1.5 suggests insufficient recovery.

46 Session Quality Scoring

A composite session quality score provides a single metric for each practice:

$$Q_{\text{session}} = w_1 \cdot A_{\text{alignment}} + w_2 \cdot S_{\text{smoothness}} + w_3 \cdot B_{\text{balance}} + w_4 \cdot C_{\text{completeness}} + w_5 \cdot P_{\text{progress}}$$

where:

- $A_{\text{alignment}}$: fraction of hold time with all joints within target ranges.
- $S_{\text{smoothness}}$: mean SPARC across all transitions.
- B_{balance} : mean bilateral symmetry across all bilateral poses.
- $C_{\text{completeness}}$: fraction of planned sequence completed.
- P_{progress} : number of poses with personal-best angle improvements.

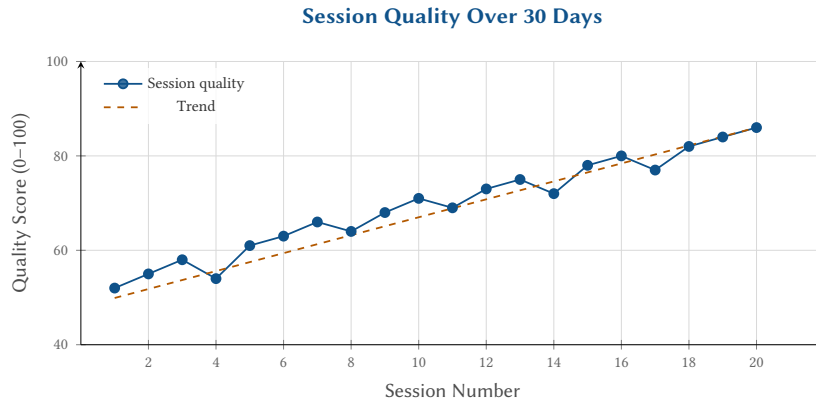


Figure 34. Session quality scores showing upward trend over 20 sessions, with natural session-to-session variability.

47 Visualisation Concepts

Rich data visualisation transforms raw analytics into actionable insights.

47.1 Body Heatmap

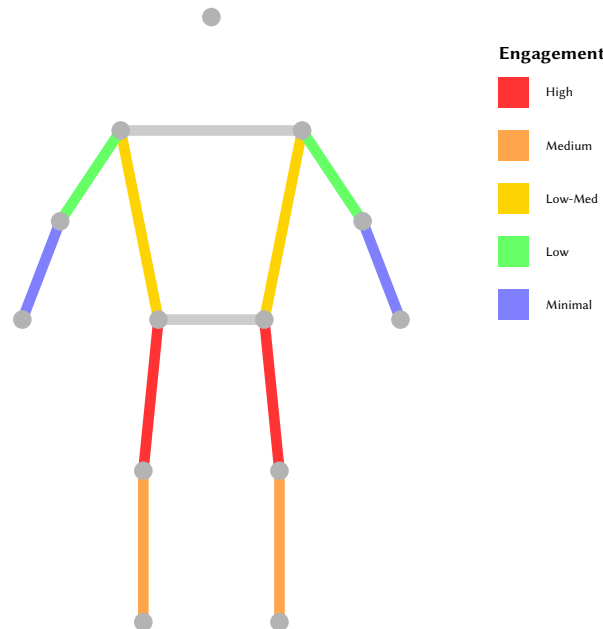


Figure 35. Body heatmap: colour-coded by engagement frequency across all sessions. Reveals imbalanced practice patterns (here: lower body dominant, upper body under-worked).

47.2 Spider Chart: Multi-Dimensional Profile

47.3 Calendar Heatmap

A GitHub-style calendar heatmap shows practice consistency at a glance, with cell colour intensity proportional to session quality score and streak indicators for consecutive practice days.

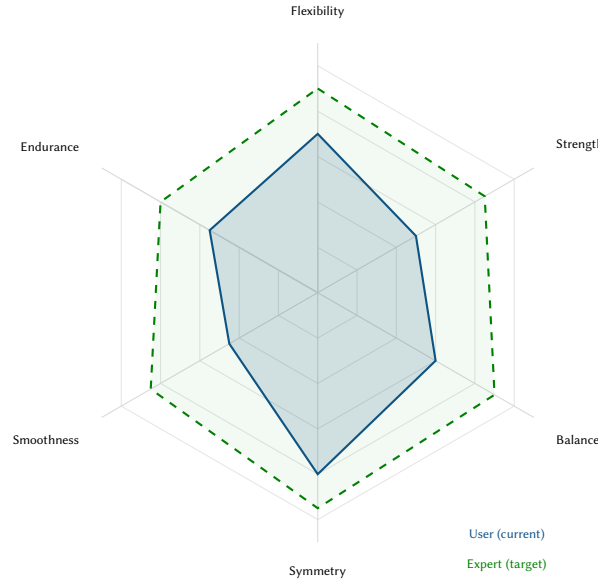


Figure 36. Spider chart comparing the user's current multi-dimensional practice profile against an expert reference.

47.4 Pose Constellation Map

Using t-SNE or UMAP dimensionality reduction on the pose embedding space, all poses in the user's repertoire are displayed as a 2D scatter plot. Explored poses are bright; unexplored poses are dim. Proximity reflects pose similarity, revealing natural clusters (standing, seated, inversions, balances) and suggesting new poses adjacent to the user's comfort zone.

48 Periodisation and Training Load

Borrowing from sports science, the system models training load to prevent overtraining and optimise progression.

48.1 Acute:Chronic Workload Ratio

The ACWR [30] compares recent training load to the long-term baseline:

$$\text{ACWR} = \frac{\bar{L}_{\text{acute}}}{\bar{L}_{\text{chronic}}}$$

where $\bar{L}_{\text{acute}}$ is the mean daily workload over the past 7 days and $\bar{L}_{\text{chronic}}$ is the mean over 28 days. The “sweet spot” for injury prevention is ACWR between 0.8 and 1.3.

Training Load and ACWR Over 8 Weeks

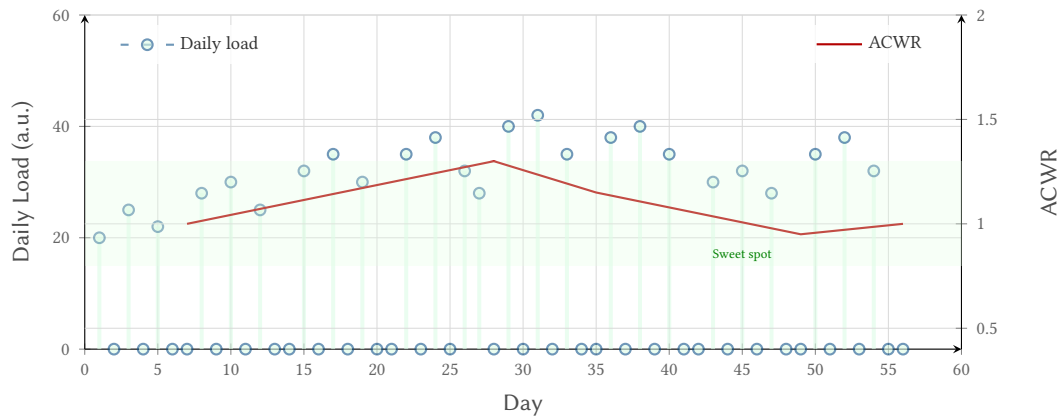


Figure 37. Training load monitoring: daily practice loads (bars) and the ACWR ratio (line). The green band marks the 0.8–1.3 sweet spot.

Design Decision 8: Training Load Calculation

Daily training load is computed as: $L = \text{duration (min)} \times \text{mean difficulty} \times \text{intensity factor}$, where intensity factor accounts for hold times, vinyasa pace, and the proportion of peak-difficulty poses. This gives a single scalar per session comparable across practice styles.

Insight:

The ACWR model enables the system to recommend rest days, suggest lighter sessions after periods of high load, and warn when training spikes risk overuse injuries—bringing sports science periodisation to yoga practice.

49 Movement Signature and Personality Profile

Over many sessions, the system identifies the user's idiosyncratic *movement vocabulary*—the shapes they naturally gravitate toward, the shapes they systematically avoid, and where their movement signature sits relative to classical archetypes.

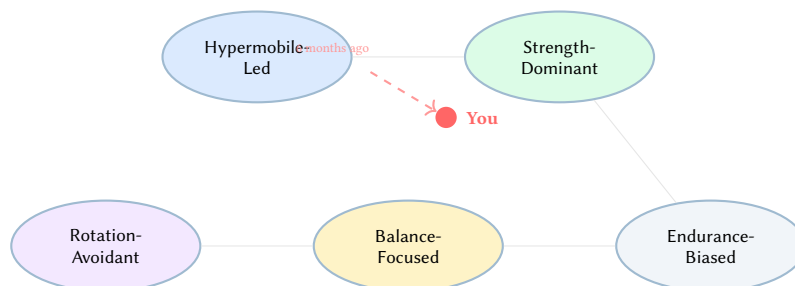


Figure 38. Movement personality map: the user's practice signature positioned among archetypal movement profiles, with a trajectory showing how their signature has evolved over months.

Archetypes are derived from UMAP clustering of the full session embedding database. The system uses the signature to generate sequences that intentionally *expand* the vocabulary (suggest under-explored movement qualities) or *reinforce* it (deepen existing strengths), depending on the user's goals.

50 Somatic Awareness and Interoception Training

Beyond external form, yoga cultivates *internal* body awareness—interoception. The system can scaffold this development.

1. **Body scan guidance with verification:** during a guided body scan, ask “bring awareness to your left hip.” Then have the user make a micro-movement at that joint. The system verifies the correct joint moved (and only that joint), confirming interoceptive accuracy.
2. **Isolation challenges:** “lift only your right shoulder without moving the left.” Score the degree of isolation (how much the non-target joints moved) as a measure of neuromuscular control.
3. **Tension mapping:** ask the user to “scan for tension” and point to areas. Compare their self-report against the system's estimate of high-torque joints (section 12). Discrepancies reveal blind spots in body awareness.
4. **Gravity awareness drill:** in a held pose, cue “where does your weight fall?” and display a foot pressure heatmap estimated from CoM projection. Compare against the user's perception.

51 Proprioceptive Calibration

Most practitioners have proprioceptive miscalibration—they *feel* level when they are actually tilted. The system can expose this gap.

1. **Blind pose test:** ask the user to close their eyes and attempt a pose they believe they know well. Compare the keypoint data against their eyes-open version.
2. **Proprioceptive drift score:** the angular difference between eyes-open and eyes-closed versions quantifies how well the user's internal body map matches reality.
3. **Targeted recalibration:** focus drills on joints with the largest proprioceptive drift—e.g., “your right hip drops 8° when you close your eyes in Warrior III.”

Insight:

Interoception training is the bridge between the physical and contemplative dimensions of yoga. It's also the hardest thing for an instructor to teach remotely—and the area where computational feedback has the highest marginal value.

Design Decision 9: Proprioceptive Training Mode

The system alternates between eyes-open and eyes-closed holds, revealing the drift in real time (after the user opens their eyes). Over weeks, the drift should decrease as proprioception improves. This is a clinically validated technique from balance rehabilitation.

52 Neuroplasticity-Informed Motor Learning

The system can apply principles from motor learning science to optimise how new poses are taught.

1. **Interleaved vs. blocked practice:** rather than drilling one pose repeatedly (blocked), alternate between 3–4 new poses (interleaved). Motor learning research shows interleaved practice produces slower initial improvement but superior long-term retention. The system can schedule this automatically.
2. **Variable practice:** introduce deliberate small variations (“try Warrior II with your feet 5 cm wider, then 5 cm narrower”) to build a richer internal model.
3. **Mental rehearsal cueing:** before a challenging pose, guide the user through a 30-second eyes-closed visualisation of the movement. Pair with the proprioceptive calibration system (section 51).
4. **Sleep-dependent consolidation:** flag poses practised late in the day as candidates for “overnight consolidation” and suggest brief review of those poses in the next morning session, leveraging sleep’s role in motor memory consolidation.
5. **Spacing effect:** schedule review of previously learned poses at expanding intervals (1 day, 3 days, 7 days, 14 days) following spaced repetition principles.

Design Decision 10: Spaced Repetition for Poses

Each pose in the user’s repertoire has a “review interval” that expands as mastery increases, identical to SRS systems like Anki. The sequence generator ensures that due-for-review poses are woven into upcoming sessions, maintaining the full repertoire without dedicated “review days.”

53 Pose Difficulty Auto-Calibration

The theoretical difficulty scores (section 20) can be refined using real-world data from the user population.

1. **Empirical difficulty:** compute the mean correction rate and mean hold-before-degradation across all users for each pose. Poses where most users struggle are empirically harder, regardless of biomechanical prediction.
2. **Personalised difficulty:** for each user, the difficulty of a pose is a function of their specific strengths and limitations. A flexible user may find Pigeon easy (low personalised difficulty) but Crow hard; a strong user may show the reverse.
3. **Difficulty trajectory:** track how a pose’s personalised difficulty changes over time. A pose that started at difficulty 8 and is now at 4 represents clear mastery progression.
4. **Difficulty-matched sequencing:** use personalised difficulty to ensure sequences are appropriately challenging (not too easy, not overwhelming), following the “zone of proximal development” principle from education.

54 Postural Carry-Over Tracking

The ultimate question: does yoga practice change how the user moves *outside* of practice?

Using phone accelerometer and gyroscope data (with consent), a companion component monitors habitual sitting and standing posture throughout the day. A *transfer score* measures how much yoga-trained alignment carries over into daily movement:

$$T_{\text{carry}} = 1 - \frac{d(\mathbf{p}_{\text{daily}}, \mathbf{p}_{\text{yoga}})}{d(\mathbf{p}_{\text{baseline}}, \mathbf{p}_{\text{yoga}})}$$

where $\mathbf{p}_{\text{daily}}$ is the mean daily posture profile, $\mathbf{p}_{\text{yoga}}$ is the trained neutral alignment, and $\mathbf{p}_{\text{baseline}}$ is the initial assessment. A transfer score approaching 1.0 means daily posture is converging on yoga-trained alignment.

Insight:
The gap between theoretical biomechanical difficulty and personalised empirical difficulty is the most actionable signal for identifying a user’s specific weaknesses and strengths.

55 Injury Recovery Journalling

When a user reports an injury, the system shifts from prevention to recovery support.

1. **Recovery timeline tracking:** log pain levels (user self-report) alongside ROM measurements at the affected joint, producing a recovery curve.
2. **Rehabilitation protocol integration:** load physiotherapist-prescribed exercises as a custom “sequence” with specific ROM targets and rep counts, tracked by the pose engine.

Insight:
Transfer score closes the loop between dedicated practice and functional daily benefit—the metric that matters most to the user but that no current yoga app attempts to measure.

3. **Return-to-practice gating:** the system blocks poses that stress the injured area until ROM and pain scores meet return-to-practice thresholds.
4. **Compensation monitoring:** during recovery, actively monitor for compensation patterns (section 24) that might develop as the body works around the injury.
5. **Practitioner–therapist sharing:** export a structured recovery report (ROM trends, compensation patterns, pain scores) as a PDF to share with a physiotherapist.

56 Meditation and Stillness Analysis

The system’s keypoint tracking works during meditation (seated or supine) to analyse stillness quality.

- **Micro-movement score:** total keypoint displacement per minute during meditation. Expert meditators show markedly less micro-movement.
- **Postural drift:** whether the head tilts or torso leans over the course of a sit, indicating loss of alertness.
- **Fidget detection:** spike detection in the movement signal identifies moments of restlessness.
- **Session arc:** a meditation quality curve showing periods of stillness vs. restlessness, helping the user identify when they settle and when they lose focus.

Meditation Stillness Profile (20-min sit)

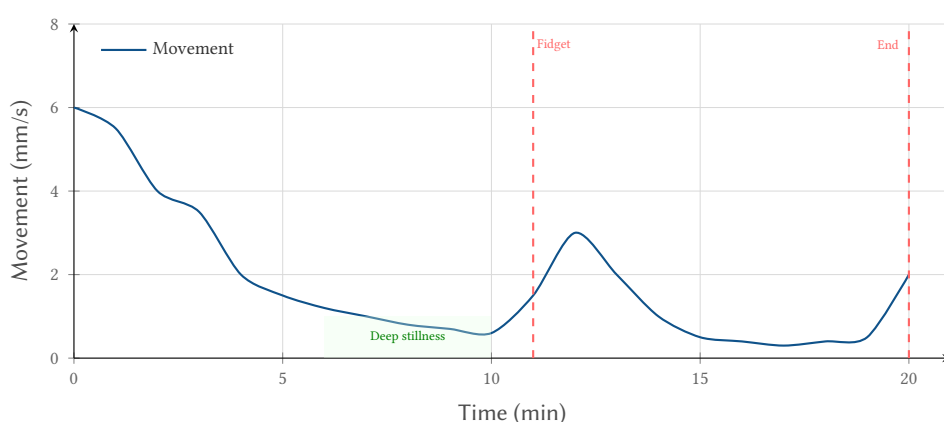


Figure 39. Meditation stillness analysis: the movement profile reveals settling time, deep stillness periods, fidget events, and end-of-sit restlessness.

57 Emotional Regulation and Nervous System Mapping

Yoga’s effects on the autonomic nervous system can be tracked and correlated with practice content.

Autonomic State Across a 45-min Practice

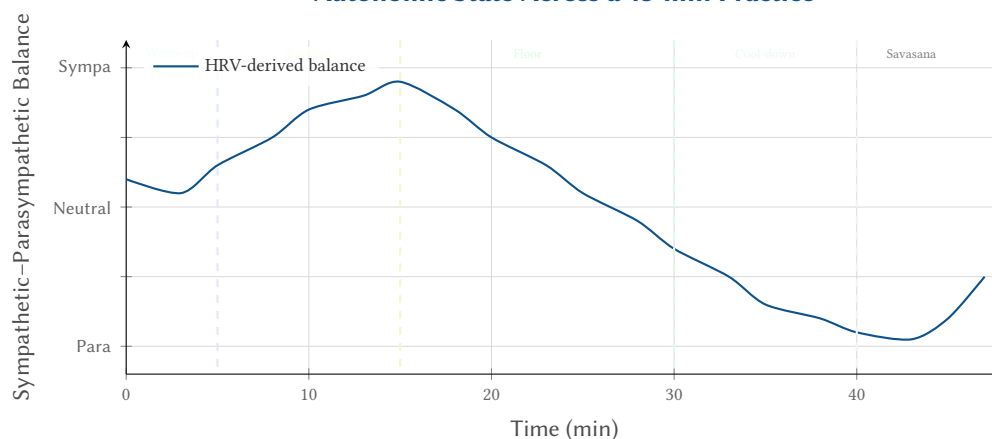


Figure 40. Autonomic nervous system balance across a full practice: sympathetic activation during vigorous standing work, parasympathetic shift during cool-down and Savasana.

- **Nervous system state tracking:** using HRV data from a wearable, classify the user’s current autonomic state (sympathetic-dominant, parasympathetic-dominant, balanced).

- **Pose–ANS correlation:** over many sessions, build a personalised map of which poses shift the user’s autonomic state in which direction. Some users find backbends energising (sympathetic); others find them calming.
- **Regulation-targeted sequences:** “I’m anxious” → generate a sequence emphasising poses that reliably shift *this user* toward parasympathetic activation, based on their personal correlation data.
- **Polyvagal mapping:** track the user’s “window of tolerance” over time—the range of autonomic arousal within which they remain regulated. Yoga practice should gradually widen this window.

Platform and Ecosystem

The preceding layers define *what* the system does; this part addresses *how* it is built and deployed. We cover the Tauri v2 application architecture, the sequence-builder graph engine, social and institutional features, accessibility, privacy-preserving analytics, and integration with wearables and the physical environment.

58 UI Framework Analysis

The application requires: live camera feed display, 2D skeleton overlay at 30 fps, information panels, a drag-and-drop sequence builder, and consumer-grade visual polish. Seven Rust UI approaches were evaluated.

Table 16. UI framework comparison for real-time pose overlay.

Framework	Camera	Overlay	Polish	Mobile	Maturity	Verdict
Tauri v2 [31]	Easy	Easy	Excellent	✓	High	Recommended
egui / eframe	Moderate	Easy	Low–Med	×	High	Best for prototyping
Iced	Moderate	Good	Med–High	×	Medium	Best pure-Rust polish
Slint	Difficult	Awkward	High	Partial	High	Poor camera fit
Dioxus (webview)	Easy	Easy	High	Exp.	Medium	Tauri-lite
Makepad	Possible	Excellent	Custom	✓	Low	High risk
Xilem	N/A	Excellent	N/A	×	V. Low	Not ready

: UI Architecture

Use Tauri v2 with a Svelte or React frontend for the UI layer, and Rust for the entire compute backend (inference, analysis, data persistence). This provides the best visual quality, easiest camera integration, and the only production-ready path to mobile (iOS/Android) in the Rust ecosystem.

Pure-Rust Alternative

If “everything in Rust” is a hard requirement, use egui for rapid prototyping or Iced for higher visual polish. Camera frames are uploaded as GPU textures each frame, with skeleton lines drawn via the framework’s 2D canvas API. Mobile is not available via this path.

59 Proposed Architecture

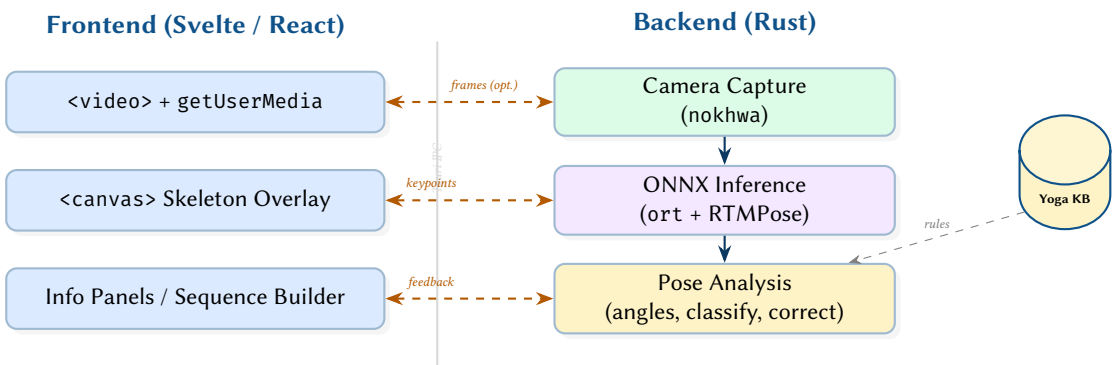


Figure 41. Proposed Tauri v2 architecture. The Rust backend handles all compute; the web frontend handles all rendering. Only keypoints and feedback text cross the IPC boundary.

Table 17. Core Rust crate stack for the backend.

Layer	Crate	Purpose
Camera capture	nokhwa 0.10.x	Cross-platform webcam
Image processing	image, imageproc	Resize, colour conversion
ML inference	ort 2.x	ONNX Runtime bindings
Linear algebra	nalgebra	Angles, Procrustes, SVD
Serialisation	serde, serde_json	IPC data transfer
Sequence matching	Custom / dtw	DTW for flow analysis
App shell	tauri 2.x	Desktop + mobile shell

59.1 Crate Dependencies

59.2 Data Flow

Two viable data flow patterns exist:

1. **Frontend-driven capture:** the browser’s `getUserMedia` feeds a `<video>` element. Frames are extracted to an off-screen `<canvas>`, transferred to Rust via Tauri commands as `base64` or `ArrayBuffer`, processed through the inference pipeline, and keypoints returned to the frontend for overlay rendering.
2. **Backend-driven capture:** Rust captures frames via `nokhwa`, runs inference, and sends only keypoints (~100 bytes/frame) to the frontend. The frontend displays a separate `<video>` from `getUserMedia` for preview and overlays the received keypoints. This avoids large frame data crossing the IPC boundary.

Design Decision 11: Recommended Data Flow

Pattern 2 (backend-driven) is recommended. Camera capture and ML inference stay in the same Rust process with zero-copy frame access. The frontend receives only lightweight keypoint data and handles rendering. Frame data never crosses the Tauri IPC boundary, which would otherwise add ~3–5 ms of serialisation/deserialisation overhead per frame at 640×480 resolution.

60 Sequence Builder as a Graph Problem

The sequence builder can be modelled as a weighted directed graph $G = (V, E, w)$ where:

- V is the set of all known poses ($|V| \approx 60$).
- $E \subseteq V \times V$ represents valid transitions.
- $w : E \rightarrow \mathbb{R}$ encodes transition quality (biomechanical compatibility, warm-up coverage, fascial line balance).

Edges are weighted by:

1. **Transition efficiency:** how naturally the exit of one pose becomes the entry of the next (shared joint positions).
2. **Warm-up coverage:** does this transition prepare the joint actions needed for subsequent poses?
3. **Fascial balance:** penalise sequences that overload one fascial line without addressing its antagonist.
4. **Contraindication penalty:** large negative weight for violating any rule in section 26.

Insight:
Sequence generation becomes shortest-path (or highest-weight path) search through the pose transition graph, constrained by the sequencing rules in section 26.

61 Generative Sequence Intelligence

Moving beyond rule-based sequence building to generative models that create novel, personalised sequences.

1. **Constraint-satisfying generation:** treat sequence creation as a constraint satisfaction problem where the constraints are the rules from section 26 (warm-up, counter-pose, no dangerous transitions), the objective is fascial line balance, and the user provides duration and intensity preferences. Solve via beam search over the pose transition graph.
2. **Style transfer:** given examples of a particular teacher’s style (their sequences as training data), generate new sequences “in the style of” that teacher.
3. **Music-synchronised flows:** generate sequences where transition timing aligns with the beat structure of a selected music track, using BPM analysis to set vinyasa pace.
4. **Progressive curriculum generation:** given a multi-week goal (e.g., “achieve Crow Pose in 8 weeks”), generate an entire curriculum of sessions with progressive difficulty, appropriate rest days, and milestone checkpoints.

5. **Adaptive mid-session replanning:** if the user struggles with a pose or skips one, replan the remainder of the sequence on-the-fly to maintain the session arc and fascial balance.

62 Wearable Integration

Smartwatch data (Apple Watch, Garmin, Fitbit) adds physiological context that the camera cannot capture.

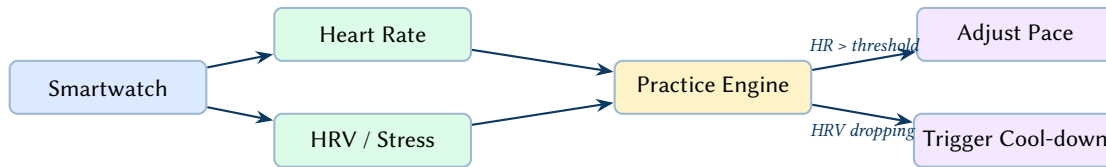


Figure 42. Wearable integration: heart rate and HRV data inform practice pacing decisions.

- **Heart rate zones:** if HR exceeds the user's aerobic threshold during a power sequence, suggest a recovery pose or longer hold in the current pose.
- **HRV-guided session length:** higher resting HRV correlates with recovery readiness; the system can recommend a more or less intense session based on pre-practice HRV.
- **Savasana biofeedback:** during final relaxation, show (or narrate) real-time HR deceleration as confirmation that the parasympathetic response is activating.
- **Breath rate estimation:** wrist-based respiration rate from PPG can cross-validate the audio/visual breath cueing tempo.

63 Ecological Biometric Integration

Combining camera-based movement analysis with non-invasive biometric signals creates a holistic practice profile.

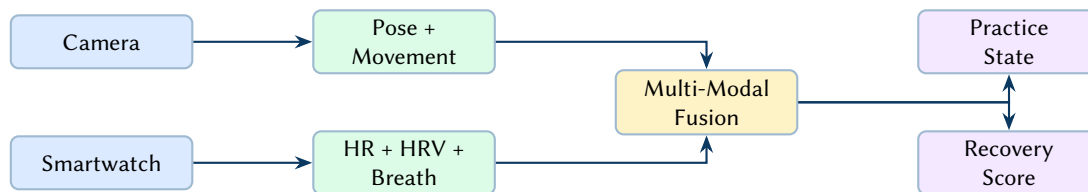


Figure 43. Multi-modal fusion: camera and wearable data are combined to estimate holistic practice state and recovery readiness.

- **Parasympathetic activation tracking:** correlate the transition from standing sequences to floor work with real-time HRV changes, validating that the cool-down is physiologically effective.
- **Flow state detection:** characterise "flow state" as the combination of low movement variability, low HR relative to effort, and consistent breath rate. Alert the user post-session when they achieved extended flow.
- **Sleep integration:** if sleep data is available (from a watch or ring), correlate evening practice content with sleep quality metrics, enabling recommendations like "your sleep quality improves 15% on days with restorative practice vs. power yoga."
- **Stress recovery protocol:** detect elevated resting HR or low morning HRV and automatically suggest a restorative or gentle sequence instead of the planned vigorous session.

64 Social and Instructor Features

64.1 Sequence Sharing

Sequences are defined as structured data (JSON or a custom DSL):

```

{
  "name": "Morning Energiser",
  "author": "Jane Doe",
  "style": "vinyasa",
  "duration_min": 25,
  "poses": [
    {"name": "mountain", "hold_s": 30, "breath": "natural"},
    {"name": "sun_sal_a", "repeats": 3},
    {"name": "warrior_1", "hold_s": 30, "side": "both"},
  ]
}
  
```

Insight:
Adaptive mid-session replanning is the key differentiator from pre-recorded video classes: the system responds to the practitioner in real time, just as a human instructor would adjust their plan based on what they observe.


```

    {"name": "warrior_2", "hold_s": 30, "side": "both"},
    {"name": "triangle", "hold_s": 25, "side": "both"},
    {"name": "tree", "hold_s": 30, "side": "both"},
    {"name": "seated_forward_fold", "hold_s": 60},
    {"name": "savasana", "hold_s": 300}
  ]
}

```

Sequences can be shared as files, QR codes, or URLs. The system validates imported sequences against the sequencing rules (section 26) and warns if contraindications are detected.

64.2 Instructor Mode

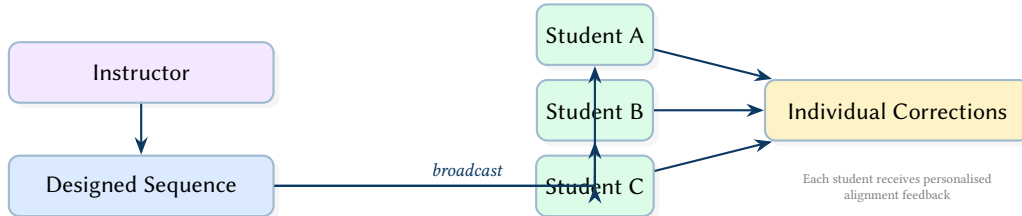


Figure 44. Instructor mode: a single sequence is broadcast to multiple students, each receiving individualised corrections from their own camera analysis.

65 Partner and Group Yoga

With multi-person pose estimation, the system can analyse partner (AcroYoga) and group dynamics.

1. **Partner alignment:** detect whether a base and flyer are correctly stacked in AcroYoga, with CoM alignment analysis and shared load distribution.
2. **Synchronisation score:** for group classes, measure how synchronised participants are in their transitions (via DTW alignment of pose sequences).
3. **Mirror mode:** in a partner context, detect whether partners are correctly mirroring each other's poses.
4. **Group difficulty balancing:** in a mixed-ability class, suggest pose modifications so that all participants are appropriately challenged.

66 Teacher Training and Institutional Mode

A B2B layer targets yoga teacher training programmes (200-hour, 300-hour RYT accreditation).

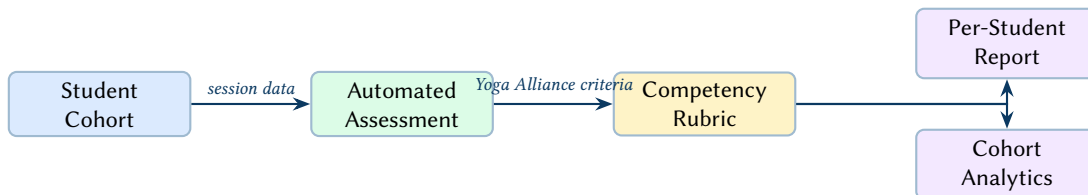


Figure 45. Teacher training mode: objective assessment reports per student against a structured competency rubric.

- **Competency tracking:** map each student's joint angle ranges, correction frequency, and consistency against a structured Yoga Alliance syllabus.
- **Objective assessment:** generate PDF reports with quantitative evidence of pose competency, replacing subjective instructor evaluation.
- **Cohort analytics:** training directors see aggregate class progress, identifying students who need extra support.
- **Teaching practice analysis:** when students teach practice classes, the system assesses *their students'* correction rates as a proxy for teaching effectiveness.

67 Cross-Discipline Movement Analysis

The pose analysis engine is fundamentally a human movement analysis system. With domain-specific rule bases, it extends to adjacent disciplines:

Table 18. Cross-discipline applications using the same inference backbone.

Discipline	Rule Base Change	Key Metric	Use Case
Pilates	Core engagement + spine neutral	Pelvic tilt angle	Mat Pilates home practice
Tai Chi	Weight shift + flow continuity	CoM trajectory	Form learning for beginners
Ballet	Turnout + port de bras alignment	Hip external rotation	Barre work at home
Martial arts	Stance width + guard position	Centre of gravity	Kata form practice
Physiotherapy	Prescribed ROM exercises	Joint angle vs. target	Rehab compliance monitoring
Ergonomics	Desk posture + screen distance	Cervical angle	Workplace health

Design Decision 12: Plugin Architecture for Disciplines

Each discipline is a plugin: a TOML file defining the pose library, correction rules, sequencing constraints, and scoring rubric. The inference backbone, UI, and analytics engine remain unchanged. This transforms the app from a yoga tool to a universal movement analysis platform.

Insight:
The most defensible moat is not the pose estimation (which is commodity ML) but the curated, expert-validated rule base for each discipline. A plugin architecture makes the knowledge base the product.

68 Privacy-Preserving Population Analytics

Users benefit from population-level comparisons without sacrificing privacy.

1. **Federated statistics:** compute local histograms of joint angle distributions per pose, inject calibrated Laplace noise (ϵ -differential privacy), and upload only the noisy histograms. The server aggregates across users to produce population percentiles.
2. **Cohort benchmarking:** users see “your hip flexion in Forward Fold is at the 62nd percentile for your age group” without any individual’s data being shared.
3. **Anonymised research corpus:** with opt-in consent, differentially private keypoint statistics contribute to a public research dataset for biomechanics and yoga science.

69 Accessibility Features

Making the system accessible to practitioners with disabilities expands impact significantly.

1. **Chair yoga mode:** adapt all pose detection and correction to seated variants, recalibrating angle thresholds and BoS calculations for a chair-based practice.
2. **Limb difference support:** gracefully handle missing limb keypoints, computing angles and corrections only for available joints, and suggesting appropriate pose modifications.
3. **Screen reader integration:** all feedback available as structured text for screen readers, not just visual overlay.
4. **High-contrast and colour-blind modes:** alternative colour palettes for the skeleton overlay and heatmaps.
5. **Deaf-friendly mode:** replace all audio cues with visual indicators, including a prominent visual metronome for breath cueing and flashing alerts for corrections.
6. **Cognitive load control:** a “simplified mode” that limits feedback to one correction at a time, with larger text and longer display durations.

: Accessibility as Core Feature

Accessibility is not an afterthought. Chair yoga mode and limb difference support should be present from Phase 3 (correction engine), as the correction rule base must be designed with these variants from the start. Retrofitting is significantly harder.

70 Movement Notation Export

Recorded sessions can be exported as formal movement notation—a human-readable score that dance teachers, physiotherapists, and researchers can interpret without video.

- **Labanotation:** the most widely used movement notation system, with a formal grammar that maps directly from joint angle time-series. Directions, levels, and durations are encoded as symbols on a vertical staff.
- **Simplified pose script:** a custom plain-text format listing pose name, entry angle, hold duration, and transition type—lightweight and version-controllable.

- **Research data export:** anonymised, timestamped keypoint CSV with per-frame joint angles, suitable for biomechanics research. Enables the app to serve as a low-cost motion capture alternative for yoga studies.

This opens cross-discipline use: sharing sequences with dance programmes, documenting clinical movement assessments, or producing reproducible research data from practice sessions.

71 Micro-Dosing Movement Throughout the Day

Instead of a single dedicated session, the system supports “movement snacks”—2–3 minute micro-practices triggered by context.

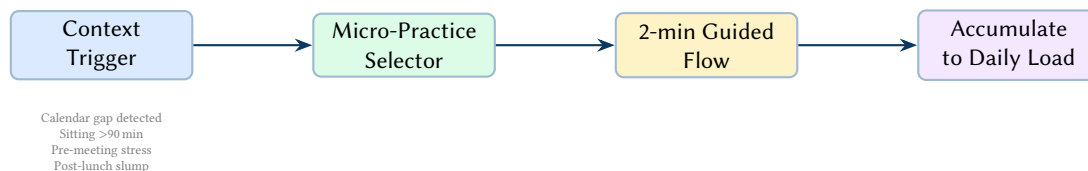


Figure 46. Micro-dosing: contextual triggers launch brief targeted movement sequences that accumulate into meaningful daily practice.

- **Desk break protocol:** 3 poses targeting hip flexors, thoracic extension, and wrist mobility—the three regions most affected by desk work.
- **Pre-meeting calm:** 2 minutes of standing poses and breath work to reduce cortisol before a stressful event.
- **Post-exercise cooldown:** detect (via calendar or wearable) that the user just ran or cycled, and offer a targeted 3-minute sport-specific stretch sequence.
- **Accumulation tracking:** sum micro-practice loads into the daily total, counting toward ACWR (section 48) and streak maintenance.

72 Spatial Flow Choreography

Most home practitioners stay rooted to one spot. The system can choreograph movement *through space*—walking transitions, directional changes, and room-scale flows.

1. **Floor space mapping:** from the initial camera calibration, estimate available floor area and obstacles.
2. **Directional transitions:** rather than always facing the camera, choreograph flows that turn 90° or 180°, simulating a studio class where the instructor calls “turn to the left wall.”
3. **Walking meditation:** track the user walking slowly across the space, analysing gait symmetry and foot placement as a mindful movement exercise.
4. **Mandala flow:** a specific vinyasa pattern where the practitioner rotates 360° around the mat over the course of a Sun Salutation cycle. The system tracks spatial position and guides rotational transitions.

Insight:
Research shows that distributed short movement bouts can match or exceed the benefits of a single long session for flexibility and pain reduction. Micro-dosing lowers the barrier to daily practice from “find 60 minutes” to “find 3 minutes, four times.”

Design Decision 13: Multi-View Spatial Tracking

With a single camera, spatial tracking relies on the user’s apparent size (scale-based depth) and skeletal orientation changes. With dual cameras (section 5), true 3D spatial tracking enables precise room-scale choreography, including tracking the user as they move in and out of each camera’s view.

73 Gamification and Motivation

Table 19. Gamification elements designed for intrinsic motivation.

Mechanic	Implementation	Motivation Type
Practice streaks	Calendar heatmap (cf. GitHub)	Consistency
Pose collection	“Unlock” new poses as prerequisites are met	Mastery
Flexibility journal	Weekly auto-generated progress snapshots	Self-awareness
Fascial explorer	Interactive body map showing explored lines	Curiosity
Challenge sequences	Community-shared themed sequences	Social
Mindfulness minutes	Cumulative time in Savasana/meditation	Wellbeing

Insight:
Gamification in a yoga context must be subtle—competition and urgency are antithetical to the practice. Focus on consistency, exploration, and self-improvement rather than scores or leaderboards.

74 Ambient and Environmental Awareness

Additional feature directions that extend the system beyond pure pose analysis:

1. **Time-of-day sequencing:** morning sequences emphasise energising backbends and standing poses; evening sequences emphasise forward folds, hip openers, and restorative poses. The system adjusts recommendations based on the current time or the user’s stated intent (“I want to wind down”).
2. **Music integration:** generate or select ambient soundscapes matched to the session arc—upbeat during the building phase, calming during cool-down, silence during Savasana. Could integrate with Spotify or Apple Music APIs, or use procedural ambient audio.
3. **Space detection:** estimate available floor space from the camera feed (using the detected body scale and visible room boundaries). Suggest compact sequences if space is limited, or expansive sequences if the user has a full studio.
4. **Practice surface detection:** detect whether the user is on a yoga mat, carpet, or hard floor, and adjust recommendations (e.g., suggest extra padding for knee-heavy sequences on hard floors).
5. **Weather/season awareness:** in colder conditions (location-based), extend the warm-up phase and favour poses that generate internal heat; in hot conditions, favour cooling forward folds and reduce hold times for strenuous poses.

CREATIVE AND FRONTIER

Finally, we look beyond the current architecture to speculative features that push the boundaries of what a movement-analysis platform could become, beginning with pose composition as a form of creative expression.

75 Pose Composition and Creative Expression

For advanced practitioners, the system supports creative exploration of non-standard poses and transitions.

1. **Freeform mode:** the system observes without classifying or correcting, instead providing a real-time biomechanical dashboard (joint angles, CoM, torques, fascial loads) that the practitioner uses as a creative tool.
2. **Pose composition:** combine elements from multiple poses (e.g., “Warrior II legs with Eagle arms”) and the system predicts the resulting biomechanical profile, difficulty score, and any safety concerns.
3. **Transition discovery:** given two poses, generate all plausible transitions (via the manifold geodesic model, section 21) and let the practitioner explore unfamiliar movement pathways.
4. **Movement improvisation scoring:** during freeform practice, score movement variety (how much of the configuration space was explored), smoothness (SPARC across the session), and novelty (distance from the user’s typical movement vocabulary).

Design Decision 14: Creative Mode as Differentiator

Most yoga apps are prescriptive (“do this pose”). Creative mode inverts the relationship: the system becomes an instrument the practitioner plays, providing real-time biomechanical feedback as they explore. This positions the app for the advanced practitioner market, which is underserved by existing tools.

76 Conclusions and Next Steps

This report has developed the architecture for *Asana*, a Rust-based yoga practice companion whose ambition extends well beyond real-time pose estimation. Across seven layers—perception, intelligence, knowledge, feedback, longitudinal analytics, platform ecosystem, and creative frontiers—we have shown that:

1. **Real-time perception is feasible.** RTMPose-m [5] via ort [7] delivers 75.3 AP at ~15 ms CPU, fitting within a 30 fps budget. Multi-camera, depth, facial-affect, and breath signals can be fused incrementally without breaking the latency contract.
2. **Biomechanical intelligence is computable from keypoints.** Joint torques, centre-of-mass tracking, fatigue indices, movement quality metrics, and pose embeddings can all be derived from 2D (or optionally 3D) keypoint streams using established biomechanical models and linear algebra.
3. **Domain knowledge is encodable.** Myofascial meridians, sequencing constraints, pose genealogy, and yoga philosophy map onto graph structures with constraint rules that a Rust backend can evaluate in real time.

4. **Rich feedback channels are architecturally compatible.** AR overlay, haptic wearables, audio cueing, soundscape generation, and LLM-powered explanation all integrate with the proposed Tauri v2 + Rust backend stack.
5. **Longitudinal intelligence adds compounding value.** Progress tracking, predictive analytics, periodisation, and movement signature analysis transform isolated sessions into a coherent practice arc.
6. **The platform scales to community and institution.** Social features, instructor mode, teacher training, accessibility, and privacy-preserving federated analytics extend the system from individual to ecosystem.
7. **Frontier features are tractable.** Creative expression, dream-sequence reconstruction, neuroplasticity-informed learning, and somatic awareness training push the boundaries but remain architecturally feasible within the proposed stack.

76.1 Recommended Development Phases

1. **Phase 1 — Proof of Concept:** Camera capture + RTMPose inference + skeleton overlay in a Tauri v2 shell. Validate the full frame-to-display pipeline at 30 fps.
2. **Phase 2 — Pose Recognition:** Build the reference angle database from Yoga-82 images. Implement kNN classification for 20–30 common poses.
3. **Phase 3 — Correction Engine:** Implement the correction rules from table 6 with temporal smoothing and priority ranking.
4. **Phase 4 — Knowledge Base:** Encode fascial lines, muscle engagement, counter-poses, and sequencing constraints. Build the sequence suggestion engine.
5. **Phase 5 — Progress & Breath:** Session recording, longitudinal tracking, breath synchronisation, voice cueing.
6. **Phase 6 — Personalisation:** Injury-aware mode, adaptive difficulty, peak pose planner.
7. **Phase 7 — AR & Rich Feedback:** Anatomical overlay modes, LLM-powered explanations, post-session summaries.
8. **Phase 8 — Social:** Sequence sharing, instructor mode, community features.
9. **Phase 9 — Mobile:** Port to iOS/Android via Tauri v2 mobile targets with wearable integration.
10. **Phase 10 — Multi-Camera:** Dual-camera support, optional depth sensing, full 3D reconstruction.

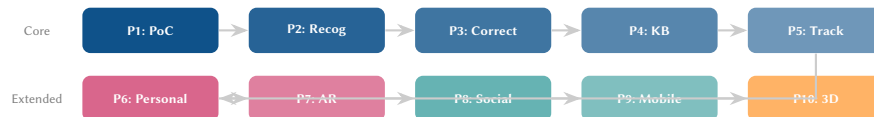


Figure 47. Ten-phase development roadmap from proof of concept to full 3D multi-camera analysis.

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Revision History

Version	Date	Changes
0.7.0	March 2026	Seven-layer reorganisation; rewritten abstract, introduction, and conclusion
0.6.0	March 2026	Somatic awareness, circadian rhythms, neuroplasticity, creative mode, recovery journalling
0.5.0	March 2026	Affect recognition, compensation detection, cross-discipline, federated analytics
0.4.0	March 2026	Digital twin, proprioception, haptics, meditation, accessibility, generative sequences
0.3.0	March 2026	Biomechanical analysis, pose embeddings, anomaly detection, practice analytics
0.2.0	March 2026	Extended features: progress tracking, breath, AR, social, LLM, wearables
0.1.0	March 2026	Initial feasibility study and architecture proposal